

NAME

BusyBox - The Swiss Army Knife of Embedded Linux

SYNTAX

```
busybox <applet> [arguments...] # or
<applet> [arguments...]         # if symlinked
```

DESCRIPTION

BusyBox combines tiny versions of many common UNIX utilities into a single small executable. It provides minimalist replacements for most of the utilities you usually find in GNU coreutils, util-linux, etc. The utilities in BusyBox generally have fewer options than their full-featured GNU cousins; however, the options that are included provide the expected functionality and behave very much like their GNU counterparts.

BusyBox has been written with size-optimization and limited resources in mind. It is also extremely modular so you can easily include or exclude commands (or features) at compile time. This makes it easy to customize your embedded systems. To create a working system, just add /dev, /etc, and a Linux kernel. BusyBox provides a fairly complete POSIX environment for any small or embedded system.

BusyBox is extremely configurable. This allows you to include only the components you need, thereby reducing binary size. Run 'make config' or 'make menuconfig' to select the functionality that you wish to enable. Then run 'make' to compile BusyBox using your configuration.

After the compile has finished, you should use 'make install' to install BusyBox. This will install the 'bin/busybox' binary, in the target directory specified by CONFIG_PREFIX. CONFIG_PREFIX can be set when configuring BusyBox, or you can specify an alternative location at install time (i.e., with a command line like 'make CONFIG_PREFIX=/tmp/foo install'). If you enabled any applet installation scheme (either as symlinks or hardlinks), these will also be installed in the location pointed to by CONFIG_PREFIX.

USAGE

BusyBox is a multi-call binary. A multi-call binary is an executable program that performs the same job as more than one utility program. That means there is just a single BusyBox binary, but that single binary acts like a large number of utilities. This allows BusyBox to be smaller since all the built-in utility programs (we call them applets) can share code for many common operations.

You can also invoke BusyBox by issuing a command as an argument on the command line. For example, entering

```
/bin/busybox ls
```

will also cause BusyBox to behave as 'ls'.

Of course, adding '/bin/busybox' into every command would be painful. So most people will invoke BusyBox using links to the BusyBox binary.

For example, entering

```
ln -s /bin/busybox ls
./ls
```

will cause BusyBox to behave as 'ls' (if the 'ls' command has been compiled into BusyBox). Generally speaking, you should never need to make all these links yourself, as the BusyBox build system will do this for you when you run the 'make install' command.

If you invoke BusyBox with no arguments, it will provide you with a list of the applets that have been compiled into your BusyBox binary.

COMMON OPTIONS

Most BusyBox applets support the **--help** argument to provide a terse runtime description of their behavior. If the CONFIG_FEATURE_VERBOSE_USAGE option has been enabled, more detailed usage information will also be available.

COMMANDS

Currently available applets include:

```
[, [, acpid, add-shell, addgroup, adduser, adjtimex, ar, arp,
arping, awk, base64, basename, bbconfig, beep, blkid, blockdev,
bootchartd, brctl, bunzip2, bzip2, cal, cat, catv, chat,
chattr, chgrp, chmod, chown, chpasswd, chpst, chroot, chrt, chvt,
cksum, clear, cmp, comm, conspy, cp, cpio, crond, crontab, cryptpw,
cttyhack, cut, date, dc, dd, deallocvt, delgroup, deluser, depmod,
devfsd, devmem, df, dhcprelay, diff, dirname, dmesg, dnsd,
dnsdomainname, dos2unix, dpkg, dpkg-deb, du, dumpkmap, dumpleases,
echo, ed, egrep, eject, env, envdir, envuidgid, ether-wake, expand,
expr, fakeidentd, false, fbset, fbsplash, fdflush, fdformat, fdisk,
fgconsole, fgrep, find, findfs, flash_eraseall, flash_lock,
flash_unlock, flashcp, flock, fold, free, freeramdisk, fsck,
fsck.minix, fsync, ftpd, ftpget, ftpput, fuser, getopt, getty, grep,
gunzip, gzip, halt, hd, hdparm, head, hexdump, hostid, hostname,
httpd, hush, hwclock, id, ifconfig, ifdown, ifenslave, ifplugd,
ifup, inetd, init, inotifyd, insmod, install, ionice, iostat, ip,
ipaddr, ipcalc, ipcrm, ipcs, iplink, iproute, iprule, iptunnel,
kbd_mode, kill, killall, killall5, klogd, last, length, less,
linux32, linux64, linuxrc, ln, loadfont, loadkmap, logger, login,
logname, logread, losetup, lpd, lpq, lpr, ls, lsattr, lsmod, lspci,
lsusb, lzcat, lzma, lzop, lzopcat, makedevs, makemime, man, md5sum,
mdev, msg, microcom, mkdir, mkdosfs, mke2fs, mkfifo, mkfs.ext2,
mkfs.minix, mkfs.reiser, mkfs.vfat, mknod, mkpasswd, mkswap, mktemp,
modinfo, modprobe, more, mount, mountpoint, mpstat, msh, mt, mv,
nameif, nanddump, nandwrite, nbd-client, nc, netstat, nice, nmeter,
nohup, nslookup, ntpd, od, openvt, passwd, patch, pgrep, pidof,
ping, ping6, pipe_progress, pivot_root, pkill, pmap, popmaildir,
poweroff, powertop, printenv, printf, ps, pscan, pwd, raidautorun,
rdate, rdev, readahead, readlink, readprofile, realpath, reboot,
reformime, remove-shell, renice, reset, resize, rev, rfc2822, rm,
rmdir, rmmmod, route, rpm, rpm2cpio, rtcwake, run-parts, runlevel,
runsv, runsvdir, rx, script, scriptreplay, sed, sendmail, seq,
setarch, setconsole, setfont, setkeycodes, setlogcons, setsid,
setuidgid, sh, shasum, sha256sum, sha512sum, showkey, slattach,
sleep, smemcap, softlimit, sort, split, start-stop-daemon, stat,
strings, stty, su, sulogin, sum, sv, svlogd, swapoff, swapon,
switch_root, sync, sysctl, syslogd, tac, tail, tar, taskset, tcpsvd,
tee, telnet, telnetd, test, tftp, tftpd, time, timeout, top, touch,
tr, traceroute, traceroute6, true, tty, ttysize, tuncctl, tune2fs,
ubiattach, ubidetach, udhcp, udhcpd, udpsvd, umount, uname,
uncompress, unexpand, uniq, unix2dos, unlzma, unlzop, unxz, unzip,
uptime, usleep, uudecode, uuencode, vconfig, vi, vlock, volname,
wall, watch, watchdog, wc, wget, which, who, whoami, xargs, xz,
xzcat, yes, zcat, zcip
```

COMMAND DESCRIPTIONS

acpid

acpid [-d] [-c CONFDIR] [-l LOGFILE] [-a ACTIONFILE] [-M MAPFILE] [-e PROC_EVENT_FILE] [-p PIDFILE]

Listen to ACPI events and spawn specific helpers on event arrival

Options:

```
-c DIR   Config directory [/etc/acpi]
-d       Don't daemonize, (implies -f)
-e FILE  /proc event file [/proc/acpi/event]
-f       Run in foreground
-l FILE  Log file [/var/log/acpid.log]
-p FILE  Pid file [/var/run/acpid.pid]
-a FILE  Action file [/etc/acpid.conf]
-M FILE  Map file [/etc/acpi.map]
```

Accept and ignore compatibility options -g -m -s -S -v

add-shell

add-shell SHELL...

Add SHELLs to /etc/shells

addgroup

addgroup [-g GID] [USER] GROUP

Add a group or add a user to a group

Options:

```
-g GID   Group id
-S       Create a system group
```

adduser

adduser [OPTIONS] USER

Add a user

Options:

```
-h DIR           Home directory
-g GECOS         GECOS field
-s SHELL         Login shell
-G GRP           Add user to existing group
-S              Create a system user
-D              Don't assign a password
-H              Don't create home directory
-u UID          User id
```

adjtimex

adjtimex [-q] [-o OFF] [-f FREQ] [-p TCONST] [-t TICK]

Read and optionally set system timebase parameters. See [adjtimex\(2\)](#).

Options:

```
-q          Quiet
-o OFF      Time offset, microseconds
-f FREQ     Frequency adjust, integer kernel units (65536 is 1ppm)
            (positive values make clock run faster)
-t TICK     Microseconds per tick, usually 10000
-p TCONST
```

ar

ar [-o] [-v] [-p] [-t] [-x] ARCHIVE FILES

Extract or list FILES from an ar archive

Options:

```
-o          Preserve original dates
-p          Extract to stdout
-t          List
-x          Extract
-v          Verbose
```

arp

arp

```
[-vn][-H HWTYPE] [-i IF] -a [HOSTNAME]
[-v] [-i IF] -d HOSTNAME [pub]
[-v] [-H HWTYPE] [-i IF] -s HOSTNAME HWADDR [temp]
[-v] [-H HWTYPE] [-i IF] -s HOSTNAME HWADDR [netmask MASK] pub
[-v] [-H HWTYPE] [-i IF] -Ds HOSTNAME IFACE [netmask MASK] pub
```

Manipulate ARP cache

Options:

-a	Display (all) hosts
-s	Set new ARP entry
-d	Delete a specified entry
-v	Verbose
-n	Don't resolve names
-i IF	Network interface
-D	Read <hwaddr> from given device
-A, -p AF	Protocol family
-H HWTYPE	Hardware address type

arping

arping [-fqbdDUA] [-c CNT] [-w TIMEOUT] [-I IFACE] [-s SRC_IP] DST_IP

Send ARP requests/replies

Options:

-f	Quit on first ARP reply
-q	Quiet
-b	Keep broadcasting, don't go unicast
-D	Duplicated address detection mode
-U	Unsolicited ARP mode, update your neighbors
-A	ARP answer mode, update your neighbors
-c N	Stop after sending N ARP requests
-w TIMEOUT	Time to wait for ARP reply, seconds
-I IFACE	Interface to use (default eth0)
-s SRC_IP	Sender IP address
DST_IP	Target IP address

awk

awk [OPTIONS] [AWK_PROGRAM] [FILE]...

Options:

-v VAR=VAL	Set variable
-F SEP	Use SEP as field separator
-f FILE	Read program from FILE

base64

base64 [-d] [FILE]

Base64 encode or decode FILE to standard output Options:

-d	Decode data
----	-------------

basename

basename FILE [SUFFIX]

Strip directory path and .SUFFIX from FILE

bbconfig

bbconfig

Print the config file used by busybox build

beep

beep -f FREQ -l LEN -d DELAY -r COUNT -n

Options:

-f	Frequency in Hz
-l	Length in ms
-d	Delay in ms
-r	Repetitions
-n	Start new tone

blkid

blkid

Print UUIDs of all filesystems

blockdev

blockdev OPTION BLOCKDEV

Options:

--setro	Set ro
--setrw	Set rw
--getro	Get ro
--getss	Get sector size
--getbsz	Get block size
--setbsz BYTES	Set block size
--getsize	Get device size in 512-byte sectors
--getsize64	Get device size in bytes
--flushbufs	Flush buffers
--rereadpt	Reread partition table

bootchartd

bootchartd start [PROG ARGS][stop|init

Create /var/log/bootchart.tgz with boot chart data

Options:

start: start background logging; with PROG, run PROG, then kill logging with USR1 stop: send USR1 to all bootchartd processes init: start background logging; stop when getty/xdm is seen (for init scripts) Under PID 1: as init, then exec \$bootchart_init, /init, /sbin/init

brctl

brctl COMMAND [BRIDGE [INTERFACE]]

Manage ethernet bridges

Commands:

show	Show a list of bridges
addbr BRIDGE	Create BRIDGE
delbr BRIDGE	Delete BRIDGE
addif BRIDGE IFACE	Add IFACE to BRIDGE
delif BRIDGE IFACE	Delete IFACE from BRIDGE
setageing BRIDGE TIME	Set ageing time
setfd BRIDGE TIME	Set bridge forward delay
sethello BRIDGE TIME	Set hello time
setmaxage BRIDGE TIME	Set max message age
setpathcost BRIDGE COST	Set path cost
setportprio BRIDGE PRIO	Set port priority
setbridgeprio BRIDGE PRIO	Set bridge priority
stp BRIDGE [1/yes/on 0/no/off]	STP on/off

bunzip2

bunzip2 [-cf] [FILE]...

Decompress FILEs (or stdin)

Options:

```
-c      Write to stdout
-f      Force
```

bzcat

bzcat FILE

Decompress to stdout

bzip2

bzip2 [OPTIONS] [FILE]...

Compress FILEs (or stdin) with bzip2 algorithm

Options:

```
-1..9   Compression level
-d      Decompress
-c      Write to stdout
-f      Force
```

cal

cal [-jy] [[MONTH] YEAR]

Display a calendar

Options:

```
-j      Use julian dates
-y      Display the entire year
```

cat

cat [FILE]...

Concatenate FILEs and print them to stdout

catv

catv [-etv] [FILE]...

Display nonprinting characters as ^x or M-x

Options:

```
-e      End each line with $
-t      Show tabs as ^I
-v      Don't use ^x or M-x escapes
```

chat

chat EXPECT [SEND [EXPECT [SEND...]]]

Useful for interacting with a modem connected to stdin/stdout. A script consists of one or more "expect-send" pairs of strings, each pair is a pair of arguments. Example: chat " ATZ OK ATD123456 CONNECT " ogin: pppuser word: ppppass '~'

chattr

chattr [-R] [-+=AacDdijsStTu] [-v VERSION] [FILE]...

Change file attributes on an ext2 fs

Modifiers:

- Remove attributes
- + Add attributes
- = Set attributes

Attributes:

- A Don't track atime
- a Append mode only
- c Enable compress
- D Write dir contents synchronously
- d Don't backup with dump
- i Cannot be modified (immutable)
- j Write all data to journal first
- s Zero disk storage when deleted
- S Write file contents synchronously
- t Disable tail-merging of partial blocks with other files
- u Allow file to be undeleted

Options:

- R Recurse
- v Set the file's version/generation number

chgrp

chgrp [-RhLHPcvf]... GROUP FILE...

Change the group membership of each FILE to GROUP

Options:

- R Recurse
- h Affect symlinks instead of symlink targets
- L Traverse all symlinks to directories
- H Traverse symlinks on command line only
- P Don't traverse symlinks (default)
- c List changed files
- v Verbose
- f Hide errors

chmod

chmod [-Rcvf] MODE[,MODE]... FILE...

Each MODE is one or more of the letters ugoa, one of the symbols += and one or more of the letters rwxst

Options:

- R Recurse
- c List changed files
- v List all files
- f Hide errors

chown

chown [-RhLHPcvf]... OWNER[<|:>[GROUP]] FILE...

Change the owner and/or group of each FILE to OWNER and/or GROUP

Options:

- R Recurse
- h Affect symlinks instead of symlink targets
- L Traverse all symlinks to directories
- H Traverse symlinks on command line only
- P Don't traverse symlinks (default)
- c List changed files
- v List all files
- f Hide errors

chpasswd

chpasswd [--md5|--encrypted]

Read user:password from stdin and update /etc/passwd

Options:

```
-e,--encrypted  Supplied passwords are in encrypted form
-m,--md5        Use MD5 encryption instead of DES
```

chpst

```
chpst [-vP012] [-u USER[:GRP]] [-U USER[:GRP]] [-e DIR]
[-/ DIR] [-n NICE] [-m BYTES] [-d BYTES] [-o N]
[-p N] [-f BYTES] [-c BYTES] PROG ARGS
```

Change the process state, run PROG

Options:

```
-u USER[:GRP]    Set uid and gid
-U USER[:GRP]    Set $UID and $GID in environment
-e DIR           Set environment variables as specified by files
                  in DIR: file=1st_line_of_file
-/ DIR           Chroot to DIR
-n NICE          Add NICE to nice value
-m BYTES         Same as -d BYTES -s BYTES -l BYTES
-d BYTES         Limit data segment
-o N             Limit number of open files per process
-p N             Limit number of processes per uid
-f BYTES         Limit output file sizes
-c BYTES         Limit core file size
-v              Verbose
-P              Create new process group
-0              Close stdin
-1              Close stdout
-2              Close stderr
```

chroot

chroot NEWROOT [PROG ARGS]

Run PROG with root directory set to NEWROOT

chrt

chrt [-prfom] [PRIO] [PID | PROG ARGS]

Change scheduling priority and class for a process

Options:

```
-p      Operate on PID
-r      Set SCHED_RR class
-f      Set SCHED_FIFO class
-o      Set SCHED_OTHER class
-m      Show min/max priorities
```

chvt

chvt N

Change the foreground virtual terminal to /dev/ttyN

cksum

cksum FILES...

Calculate the CRC32 checksums of FILES

clear

clear

Clear screen

cmp

```
cmp [-l] [-s] FILE1 [FILE2 [SKIP1 [SKIP2]]]
```


Compare FILE1 with FILE2 (or stdin)

Options:

- l Write the byte numbers (decimal) and values (octal) for all differing bytes
- s Quiet

comm

comm [-123] FILE1 FILE2

Compare FILE1 with FILE2

Options:

- 1 Suppress lines unique to FILE1
- 2 Suppress lines unique to FILE2
- 3 Suppress lines common to both files

conspy

conspy [-vcsndf] [-x COL] [-y LINE] [CONSOLE_NO]

A text-mode VNC like program for Linux virtual consoles. To exit, quickly press ESC 3 times.

Options:

- v Don't send keystrokes to the console
- c Create missing devices in /dev
- s Open a SHELL session
- n Black & white
- d Dump console to stdout
- f Follow cursor
- x COL Starting column
- y LINE Starting line

cp

cp [OPTIONS] SOURCE DEST

Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY

Options:

- a Same as -dpR
- R,-r Recurse
- d,-P Preserve symlinks (default if -R)
- L Follow all symlinks
- H Follow symlinks on command line
- p Preserve file attributes if possible
- f Overwrite
- i Prompt before overwrite
- l,-s Create (sym)links

cpio

cpio [-dmvu] [-F FILE] [-H newc] [-tio] [-p DIR]

Extract or list files from a cpio archive, or create an archive (-o) or copy files (-p) using file list on stdin

Main operation mode:

- t List
- i Extract
- o Create (requires -H newc)
- p DIR Copy files to DIR

Options:

- d Make leading directories
- m Preserve mtime
- v Verbose
- u Overwrite
- F FILE Input (-t,-i,-p) or output (-o) file
- H newc Archive format

crond

crond -fbS -l N -d N -L LOGFILE -c DIR

```
-f      Foreground
-b      Background (default)
-S      Log to syslog (default)
-l      Set log level. 0 is the most verbose, default 8
-d      Set log level, log to stderr
-L      Log to file
-c      Working dir
```

crontab

crontab [-c DIR] [-u USER] [-ler][FILE]

```
-c      Crontab directory
-u      User
-l      List crontab
-e      Edit crontab
-r      Delete crontab
FILE    Replace crontab by FILE ('-' : stdin)
```

cryptpw

cryptpw [OPTIONS] [PASSWORD] [SALT]

Crypt the PASSWORD using crypt(3)

Options:

```
-P,--password-fd=N    Read password from fd N
-m,--method=TYPE      Encryption method TYPE
-S,--salt=SALT
```

cttyhack

cttyhack PROG ARGS

Give PROG a controlling tty if possible. Example for /etc/inittab (for busybox init): ::respawn:/bin/cttyhack /bin/sh
Giving controlling tty to shell running with PID 1: \$ exec cttyhack sh Starting interactive shell from boot shell script:

```
setsid cttyhack sh
```

cut

cut [OPTIONS] [FILE]...

Print selected fields from each input FILE to stdout

Options:

```
-b LIST Output only bytes from LIST
-c LIST Output only characters from LIST
-d CHAR Use CHAR instead of tab as the field delimiter
-s      Output only the lines containing delimiter
-f N    Print only these fields
-n      Ignored
```

date

date [OPTIONS] [+FMT] [TIME]

Display time (using +FMT), or set time

Options:

```
[-s,--set] TIME Set time to TIME
-u,--utc        Work in UTC (don't convert to local time)
-R,--rfc-2822   Output RFC-2822 compliant date string
-I[SPEC]        Output ISO-8601 compliant date string
                 SPEC='date' (default) for date only,
                 'hours', 'minutes', or 'seconds' for date and
                 time to the indicated precision
```

```
-r,--reference FILE      Display last modification time of FILE
-d,--date TIME          Display TIME, not 'now'
-D FMT                  Use FMT for -d TIME conversion
```

Recognized TIME formats:

```
hh:mm[:ss]
[YYYY.]MM.DD-hh:mm[:ss]
YYYY-MM-DD hh:mm[:ss]
[[[YY]YY]MM]DD]hh]mm[:ss]
```

dc

dc EXPRESSION...

Tiny RPN calculator. Operations: +, add, -, sub, *, mul, /, div, %, mod, **, exp, and, or, not, eor, p - print top of the stack (without popping), f - print entire stack, o - pop the value and set output radix (must be 10, 16, 8 or 2).

Examples: 'dc 2 2 add' -> 4, 'dc 8 8 * 2 2 + /' -> 16

dd

```
dd [if=FILE] [of=FILE] [ibs=N] [obs=N] [bs=N] [count=N] [skip=N]
[seek=N] [conv=notrunc|noerror|sync|fsync]
```

Copy a file with converting and formatting

Options:

```
if=FILE      Read from FILE instead of stdin
of=FILE      Write to FILE instead of stdout
bs=N         Read and write N bytes at a time
ibs=N        Read N bytes at a time
obs=N        Write N bytes at a time
count=N      Copy only N input blocks
skip=N       Skip N input blocks
seek=N       Skip N output blocks
conv=notrunc Don't truncate output file
conv=noerror Continue after read errors
conv=sync    Pad blocks with zeros
conv=fsync   Physically write data out before finishing
```

Numbers may be suffixed by c (x1), w (x2), b (x512), kD (x1000), k (x1024), MD (x1000000), M (x1048576), GD (x1000000000) or G (x1073741824)

deallocvt

deallocvt [N]

Deallocate unused virtual terminal /dev/ttyN

delgroup

delgroup [USER] GROUP

Delete group GROUP from the system or user USER from group GROUP

deluser

deluser USER

Delete USER from the system

depmod

depmod [-qfwrsv] MODULE [symbol=value]...

Options:

```
-r      Remove MODULE (stacks) or do autoclean
-q      Quiet
-v      Verbose
-f      Force
```

```
-w      Wait for unload
-s      Report via syslog instead of stderr
```

devfsd

```
devfsd mntpnt [-v][-fg][-np]
```

Manage devfs permissions and old device name symlinks

Options:

```
mntpnt  The mount point where devfs is mounted
-v      Print the protocol version numbers for devfsd
        and the kernel-side protocol version and exit
-fg     Run in foreground
-np     Exit after parsing the configuration file
        and processing synthetic REGISTER events,
        don't poll for events
```

devmem

```
devmem ADDRESS [WIDTH [VALUE]]
```

Read/write from physical address

```
ADDRESS Address to act upon
WIDTH   Width (8/16/...)
VALUE   Data to be written
```

df

```
df [-Pkmhai] [-B SIZE] [FILESYSTEM]...
```

Print filesystem usage statistics

Options:

```
-P      POSIX output format
-k      1024-byte blocks (default)
-m      1M-byte blocks
-h      Human readable (e.g. 1K 243M 2G)
-a      Show all filesystems
-i      Inodes
-B SIZE Blocksize
```

dhcprelay

```
dhcprelay CLIENT_IFACE[,CLIENT_IFACE2]... SERVER_IFACE [SERVER_IP]
```

Relay DHCP requests between clients and server

diff

```
diff [-abBdiNqrTstw] [-L LABEL] [-S FILE] [-U LINES] FILE1 FILE2
```

Compare files line by line and output the differences between them. This implementation supports unified diffs only.

Options:

```
-a      Treat all files as text
-b      Ignore changes in the amount of whitespace
-B      Ignore changes whose lines are all blank
-d      Try hard to find a smaller set of changes
-i      Ignore case differences
-L      Use LABEL instead of the filename in the unified header
-N      Treat absent files as empty
-q      Output only whether files differ
-r      Recurse
-S      Start with FILE when comparing directories
-T      Make tabs line up by prefixing a tab when necessary
-s      Report when two files are the same
-t      Expand tabs to spaces in output
-U      Output LINES lines of context
-w      Ignore all whitespace
```

dirname

dirname FILENAME

Strip non-directory suffix from FILENAME

dmesg

dmesg [-c] [-n LEVEL] [-s SIZE]

Print or control the kernel ring buffer

Options:

-c	Clear ring buffer after printing
-n LEVEL	Set console logging level
-s SIZE	Buffer size

dnssd

dnssd [-dvs] [-c CONFFILE] [-t TTL_SEC] [-p PORT] [-i ADDR]

Small static DNS server daemon

Options:

-c FILE	Config file
-t SEC	TTL
-p PORT	Listen on PORT
-i ADDR	Listen on ADDR
-d	Daemonize
-v	Verbose
-s	Send successful replies only. Use this if you want to use /etc/resolv.conf with two nameserver lines: nameserver DNSD_SERVER nameserver NORMAL_DNS_SERVER

dos2unix

dos2unix [-ud] [FILE]

Convert FILE in-place from DOS to Unix format. When no file is given, use stdin/stdout.

Options:

-u	dos2unix
-d	unix2dos

dpkg

dpkg [-ilCPru] [-F OPT] PACKAGE

Install, remove and manage Debian packages

Options:

-i,--install	Install the package
-l,--list	List of installed packages
--configure	Configure an unpackaged package
-P,--purge	Purge all files of a package
-r,--remove	Remove all but the configuration files for a package
--unpack	Unpack a package, but don't configure it
--force-depends	Ignore dependency problems
--force-confnew	Overwrite existing config files when installing
--force-confold	Keep old config files when installing

dpkg-deb

dpkg-deb [-cefxX] FILE [argument]

Perform actions on Debian packages (.debs)

Options:

```
-c      List contents of filesystem tree
-e      Extract control files to [argument] directory
-f      Display control field name starting with [argument]
-x      Extract packages filesystem tree to directory
-X      Verbose extract
```

du

`du [-aHLdclsxhmk] [FILE]...`

Summarize disk space used for each FILE and/or directory. Disk space is printed in units of 1024 bytes.

Options:

```
-a      Show file sizes too
-L      Follow all symlinks
-H      Follow symlinks on command line
-d N    Limit output to directories (and files with -a) of depth < N
-c      Show grand total
-l      Count sizes many times if hard linked
-s      Display only a total for each argument
-x      Skip directories on different filesystems
-h      Sizes in human readable format (e.g., 1K 243M 2G )
-m      Sizes in megabytes
-k      Sizes in kilobytes (default)
```

dumpkmap

`dumpkmap > keymap`

Print a binary keyboard translation table to stdout

dumpleases

`dumpleases [-r|-a] [-f LEASEFILE]`

Display DHCP leases granted by udhcpd

Options:

```
-f,--file=FILE  Lease file
-r,--remaining  Show remaining time
-a,--absolute   Show expiration time
```

echo

`echo [-neE] [ARG]...`

Print the specified ARGs to stdout

Options:

```
-n      Suppress trailing newline
-e      Interpret backslash escapes (i.e., \t=tab)
-E      Don't interpret backslash escapes (default)
```

ed

`ed`

eject

`eject [-t] [-T] [DEVICE]`

Eject DEVICE or default /dev/cdrom

Options:

```
-s      SCSI device
-t      Close tray
-T      Open/close tray (toggle)
```

env

env [-iu] [-] [name=value]... [PROG ARGS]

Print the current environment or run PROG after setting up the specified environment

Options:

- , -i Start with an empty environment
 -u Remove variable from the environment

envdir

envdir DIR PROG ARGS

Set various environment variables as specified by files in the directory DIR, run PROG

envuidgid

envuidgid USER PROG ARGS

Set \$UID to USER's uid and \$GID to USER's gid, run PROG

ether-wake

ether-wake [-b] [-i iface] [-p aa:bb:cc:dd[:ee:ff]] MAC

Send a magic packet to wake up sleeping machines. MAC must be a station address (00:11:22:33:44:55) or a hostname with a known 'ethers' entry.

Options:

-b Send wake-up packet to the broadcast address
 -i iface Interface to use (default eth0)
 -p pass Append four or six byte password PW to the packet

expand

expand [-i] [-t N] [FILE]...

Convert tabs to spaces, writing to stdout

Options:

-i,--initial Don't convert tabs after non blanks
 -t,--tabs=N Tabstops every N chars

expr

expr EXPRESSION

Print the value of EXPRESSION to stdout

EXPRESSION may be:

ARG1 ARG2	ARG1 if it is neither null nor 0, otherwise ARG2
ARG1 & ARG2	ARG1 if neither argument is null or 0, otherwise 0
ARG1 < ARG2	1 if ARG1 is less than ARG2, else 0. Similarly:
ARG1 <= ARG2	
ARG1 = ARG2	
ARG1 != ARG2	
ARG1 >= ARG2	
ARG1 > ARG2	
ARG1 + ARG2	Sum of ARG1 and ARG2. Similarly:
ARG1 - ARG2	
ARG1 * ARG2	
ARG1 / ARG2	
ARG1 % ARG2	
STRING : REGEXP	Anchored pattern match of REGEXP in STRING
match STRING REGEXP	Same as STRING : REGEXP
substr STRING POS LENGTH	Substring of STRING, POS counted from 1
index STRING CHARS	Index in STRING where any CHARS is found, or 0
length STRING	Length of STRING
quote TOKEN	Interpret TOKEN as a string, even if it is a keyword like 'match' or an

(EXPRESSION) operator like '/'
Value of EXPRESSION

Beware that many operators need to be escaped or quoted for shells. Comparisons are arithmetic if both ARGs are numbers, else lexicographical. Pattern matches return the string matched between \(and\) or null; if \(and\) are not used, they return the number of characters matched or 0.

fakeidentd

fakeidentd [-fiw] [-b ADDR] [STRING]

Provide fake ident (auth) service

Options:

```
-f      Run in foreground
-i      Inetd mode
-w      Inetd 'wait' mode
-b ADDR Bind to specified address
STRING Ident answer string (default: nobody)
```

false

false

Return an exit code of FALSE (1)

fbset

fbset [OPTIONS] [MODE]

Show and modify frame buffer settings

fbsplash

fbsplash -s IMGFILE [-c] [-d DEV] [-i INIFILE] [-f CMD]

Options:

```
-s      Image
-c      Hide cursor
-d      Framebuffer device (default /dev/fb0)
-i      Config file (var=value):
        BAR_LEFT,BAR_TOP,BAR_WIDTH,BAR_HEIGHT
        BAR_R,BAR_G,BAR_B
-f      Control pipe (else exit after drawing image)
        commands: 'NN' (% for progress bar) or 'exit'
```

fdflush

fdflush DEVICE

Force floppy disk drive to detect disk change

fdformat

fdformat [-n] DEVICE

Format floppy disk

Options:

```
-n      Don't verify after format
```

fdisk

fdisk [-ul] [-C CYLINDERS] [-H HEADS] [-S SECTORS] [-b SSZ] DISK

Change partition table

Options:

-u	Start and End are in sectors (instead of cylinders)
-l	Show partition table for each DISK, then exit
-b 2048	(for certain M0 disks) use 2048-byte sectors
-C CYLINDERS	Set number of cylinders/heads/sectors
-H HEADS	
-S SECTORS	

fgconsole

fgconsole

Get active console

find

find [PATH]... [EXPRESSION]

Search for files. The default PATH is the current directory, default EXPRESSION is '-print'

EXPRESSION may consist of:

-follow	Follow symlinks
-xdev	Don't descend directories on other filesystems
-maxdepth N	Descend at most N levels. -maxdepth 0 applies tests/actions to command line arguments only
-mindepth N	Don't act on first N levels
-name PATTERN	File name (w/o directory name) matches PATTERN
-iname PATTERN	Case insensitive -name
-path PATTERN	Path matches PATTERN
-regex PATTERN	Path matches regex PATTERN
-type X	File type is X (X is one of: f,d,l,b,c,...)
-perm NNN	Permissions match any of (+NNN), all of (-NNN), or exactly NNN
-mtime DAYS	Modified time is greater than (+N), less than (-N), or exactly N days
-mmin MINS	Modified time is greater than (+N), less than (-N), or exactly N minutes
-newer FILE	Modified time is more recent than FILE's
-inum N	File has inode number N
-user NAME	File is owned by user NAME (numeric user ID allowed)
-group NAME	File belongs to group NAME (numeric group ID allowed)
-depth	Process directory name after traversing it
-size N[bck]	File size is N (c:bytes,k:kbytes,b:512 bytes(def.))
	+/-N: file size is bigger/smaller than N
-links N	Number of links is greater than (+N), less than (-N), or exactly N
-print	Print (default and assumed)
-print0	Delimit output with null characters rather than newlines
-exec CMD ARG ;	Run CMD with all instances of {} replaced by the matching files
-prune	Stop traversing current subtree
-delete	Delete files, turns on -depth option
(EXPR)	Group an expression

findfs

findfs LABEL=label or UUID=uuid

Find a filesystem device based on a label or UUID

flash_eraseall

flash_eraseall [-jq] MTD_DEVICE

Erase an MTD device

Options:

-j	Format the device for jffs2
-q	Don't display progress messages

flash_lock

flash_lock MTD_DEVICE OFFSET SECTORS

Lock part or all of an MTD device. If SECTORS is -1, then all sectors will be locked, regardless of the value of OFFSET

flash_unlock

flash_unlock MTD_DEVICE

Unlock an MTD device

flashcp

flashcp -v FILE MTD_DEVICE

Copy an image to MTD device

Options:

-v Verbose

flock

flock [-sxun] FD[{FILE [-c] PROG ARGS}]

[Un]lock file descriptor, or lock FILE, run PROG

Options:

-s Shared lock
-x Exclusive lock (default)
-u Unlock FD
-n Fail rather than wait

fold

fold [-bs] [-w WIDTH] [FILE]...

Wrap input lines in each FILE (or stdin), writing to stdout

Options:

-b Count bytes rather than columns
-s Break at spaces
-w Use WIDTH columns instead of 80

free

free [-b/k/m/g]

Display the amount of free and used system memory

freeramdisk

freeramdisk DEVICE

Free all memory used by the specified ramdisk

fsck

fsck [-ANPRTV] [-C FD] [-t FSTYPE] [FS_OPTS] [BLOCKDEV]...

Check and repair filesystems

Options:

-A Walk /etc/fstab and check all filesystems
-N Don't execute, just show what would be done
-P With -A, check filesystems in parallel
-R With -A, skip the root filesystem
-T Don't show title on startup
-V Verbose
-C n Write status information to specified filedescriptor
-t TYPE List of filesystem types to check

fsck.minix

fsck.minix [-larvsmf] BLOCKDEV

Check MINIX filesystem

Options:

-l	List all filenames
-r	Perform interactive repairs
-a	Perform automatic repairs
-v	Verbose
-s	Output superblock information
-m	Show "mode not cleared" warnings
-f	Force file system check

fsync

fsync [-d] FILE...

Write files' buffered blocks to disk

Options:

-d	Avoid syncing metadata
----	------------------------

ftpd

ftpd [-wvS] [-t N] [-T N] [DIR]

Anonymous FTP server

ftpd should be used as an inetd service. ftpd's line for inetd.conf: 21 stream tcp nowait root ftpd ftpd /files/to/serve It also can be ran from tcpsvd:

```
tcpsvd -vE 0.0.0.0 21 ftpd /files/to/serve
```

Options:

-w	Allow upload
-v	Log to stderr
-S	Log to syslog
-t, -T	Idle and absolute timeouts
DIR	Change root to this directory

ftpget

ftpget [OPTIONS] HOST [LOCAL_FILE] REMOTE_FILE

Retrieve a remote file via FTP

Options:

-c, --continue	Continue previous transfer
-v, --verbose	Verbose
-u, --username	Username
-p, --password	Password
-P, --port	Port number

ftpput

ftpput [OPTIONS] HOST [REMOTE_FILE] LOCAL_FILE

Store a local file on a remote machine via FTP

Options:

-v, --verbose	Verbose
-u, --username	Username
-p, --password	Password
-P, --port	Port number

fuser

fuser [OPTIONS] FILE or PORT/PROTO

Find processes which use FILEs or PORTs

Options:

```
-m      Find processes which use same fs as FILEs
-4,-6   Search only IPv4/IPv6 space
-s      Don't display PIDs
-k      Kill found processes
-SIGNAL Signal to send (default: KILL)
```

getopt

getopt [OPTIONS]

Options:

```
-a,--alternative      Allow long options starting with single -
-l,--longoptions=longopts  Long options to be recognized
-n,--name=progname     The name under which errors are reported
-o,--options=optstring  Short options to be recognized
-q,--quiet             Disable error reporting by getopt(3)
-Q,--quiet-output      No normal output
-s,--shell=shell        Set shell quoting conventions
-T,--test              Test for getopt(1) version
-u,--unquoted          Don't quote the output
```

getty

getty [OPTIONS] BAUD_RATE TTY [TERMTYPE]

Open a tty, prompt for a login name, then invoke /bin/login

Options:

```
-h      Enable hardware (RTS/CTS) flow control
-i      Don't display /etc/issue before running login
-L      Local line, don't do carrier detect
-m      Get baud rate from modem's CONNECT status message
-w      Wait for a CR or LF before sending /etc/issue
-n      Don't prompt the user for a login name
-f ISSUE_FILE  Display ISSUE_FILE instead of /etc/issue
-l LOGIN      Invoke LOGIN instead of /bin/login
-t SEC      Terminate after SEC if no username is read
-I INITSTR    Send INITSTR before anything else
-H HOST      Log HOST into the utmp file as the hostname
```

grep

grep [-HhnlLoqvsriwFEz] [-m N] [-A/B/C N] PATTERN/-e PATTERN.../-f FILE [FILE]...

Search for PATTERN in FILEs (or stdin)

Options:

```
-H      Add 'filename:' prefix
-h      Do not add 'filename:' prefix
-n      Add 'line_no:' prefix
-l      Show only names of files that match
-L      Show only names of files that don't match
-c      Show only count of matching lines
-o      Show only the matching part of line
-q      Quiet. Return 0 if PATTERN is found, 1 otherwise
-v      Select non-matching lines
-s      Suppress open and read errors
-r      Recurse
-i      Ignore case
-w      Match whole words only
-F      PATTERN is a literal (not regexp)
-E      PATTERN is an extended regexp
-z      Input is NUL terminated
-m N    Match up to N times per file
-A N    Print N lines of trailing context
-B N    Print N lines of leading context
```

```
-C N    Same as '-A N -B N'
-e PTRN Pattern to match
-f FILE Read pattern from file
```

gunzip

gunzip [-cft] [FILE]...

Decompress FILEs (or stdin)

Options:

```
-c      Write to stdout
-f      Force
-t      Test file integrity
```

gzip

gzip [-cfd] [FILE]...

Compress FILEs (or stdin)

Options:

```
-d      Decompress
-c      Write to stdout
-f      Force
```

halt

halt [-d DELAY] [-n] [-f] [-w]

Halt the system

Options:

```
-d SEC  Delay interval
-n      Do not sync
-f      Force (don't go through init)
-w      Only write a wtmp record
```

hd

hd FILE...

hd is an alias for hexdump -C

hdparm

hdparm [OPTIONS] [DEVICE]

Options:

```
-a      Get/set fs readahead
-A      Set drive read-lookahead flag (0/1)
-b      Get/set bus state (0 == off, 1 == on, 2 == tristate)
-B      Set Advanced Power Management setting (1-255)
-c      Get/set IDE 32-bit I0 setting
-C      Check IDE power mode status
-d      Get/set using_dma flag
-D      Enable/disable drive defect-mgmt
-f      Flush buffer cache for device on exit
-g      Display drive geometry
-h      Display terse usage information
-i      Display drive identification
-I      Detailed/current information directly from drive
-k      Get/set keep_settings_over_reset flag (0/1)
-K      Set drive keep_features_over_reset flag (0/1)
-L      Set drive doorlock (0/1) (removable harddisks only)
-m      Get/set multiple sector count
-n      Get/set ignore-write-errors flag (0/1)
-p      Set PIO mode on IDE interface chipset (0,1,2,3,4,...)
-P      Set drive prefetch count
-Q      Get/set DMA tagged-queuing depth (if supported)
```

```

-r      Get/set readonly flag (DANGEROUS to set)
-R      Register an IDE interface (DANGEROUS)
-S      Set standby (spindown) timeout
-t      Perform device read timings
-T      Perform cache read timings
-u      Get/set unmaskirq flag (0/1)
-U      Unregister an IDE interface (DANGEROUS)
-v      Defaults; same as -mcudkrag for IDE drives
-V      Display program version and exit immediately
-w      Perform device reset (DANGEROUS)
-W      Set drive write-caching flag (0/1) (DANGEROUS)
-x      Tristate device for hotswap (0/1) (DANGEROUS)
-X      Set IDE xfer mode (DANGEROUS)
-y      Put IDE drive in standby mode
-Y      Put IDE drive to sleep
-Z      Disable Seagate auto-powersaving mode
-z      Reread partition table

```

head

head [OPTIONS] [FILE]...

Print first 10 lines of each FILE (or stdin) to stdout. With more than one FILE, precede each with a filename header.

Options:

```

-n N[kbm]      Print first N lines
-c N[kbm]      Print first N bytes
-q             Never print headers
-v             Always print headers

```

N may be suffixed by k (x1024), b (x512), or m (x1024^2).

hexdump

hexdump [-bcCdefnosvxR] [FILE]...

Display FILEs (or stdin) in a user specified format

Options:

```

-b          One-byte octal display
-c          One-byte character display
-C          Canonical hex+ASCII, 16 bytes per line
-d          Two-byte decimal display
-e FORMAT STRING
-f FORMAT FILE
-n LENGTH   Interpret only LENGTH bytes of input
-o          Two-byte octal display
-s OFFSET   Skip OFFSET bytes
-v          Display all input data
-x          Two-byte hexadecimal display
-R          Reverse of 'hexdump -Cv'

```

hostid

hostid

Print out a unique 32-bit identifier for the machine

hostname

hostname [OPTIONS] [HOSTNAME | -F FILE]

Get or set hostname or DNS domain name

Options:

```

-s          Short
-i          Addresses for the hostname
-d          DNS domain name
-f          Fully qualified domain name
-F FILE     Use FILE's content as hostname

```

httpd

httpd [-ifv[v]] [-c CONFFILE] [-p [IP:]PORT] [-u USER[:GRP]] [-r REALM] [-h HOME] or httpd -d/-e/-m STRING

Listen for incoming HTTP requests

Options:

```
-i          Inetd mode
-f          Don't daemonize
-v[v]      Verbose
-p [IP:]PORT Bind to IP:PORT (default *:80)
-u USER[:GRP] Set uid/gid after binding to port
-r REALM    Authentication Realm for Basic Authentication
-h HOME     Home directory (default .)
-c FILE     Configuration file (default {/etc,HOME}/httpd.conf)
-m STRING   MD5 crypt STRING
-e STRING   HTML encode STRING
-d STRING   URL decode STRING
```

hwclock

hwclock [-r|--show] [-s|--hctosys] [-w|--systohc] [-l|--localtime] [-u|--utc] [-f FILE]

Query and set hardware clock (RTC)

Options:

```
-r          Show hardware clock time
-s          Set system time from hardware clock
-w          Set hardware clock to system time
-u          Hardware clock is in UTC
-l          Hardware clock is in local time
-f FILE     Use specified device (e.g. /dev/rtc2)
```

id

id [OPTIONS] [USER]

Print information about USER or the current user

Options:

```
-u          User ID
-g          Group ID
-G          Supplementary group IDs
-n          Print names instead of numbers
-r          Print real ID instead of effective ID
```

ifconfig

ifconfig [-a] interface [address]

Configure a network interface

Options:

```
[add ADDRESS[/PREFIXLEN]]
[del ADDRESS[/PREFIXLEN]]
[[-]broadcast [ADDRESS]] [[-]pointopoint [ADDRESS]]
[netmask ADDRESS] [dstaddr ADDRESS]
[outfill NN] [keepalive NN]
[hw ether|infiniband ADDRESS] [metric NN] [mtu NN]
[[-]trailers] [[-]arp] [[-]allmulti]
[multicast] [[-]promisc] [txqueuelen NN] [[-]dynamic]
[mem_start NN] [io_addr NN] [irq NN]
[up|down] ...
```

ifdown

ifdown [-anmvf] [-i FILE] IFACE...

Options:

```
-a          De/configure all interfaces automatically
-i FILE     Use FILE for interface definitions
```

```
-n      Print out what would happen, but don't do it
        (note: doesn't disable mappings)
-m      Don't run any mappings
-v      Print out what would happen before doing it
-f      Force de/configuration
```

ifenslave

ifenslave [-cdf] MASTER_IFACE SLAVE_IFACE...

Configure network interfaces for parallel routing

Options:

```
-c,--change-active  Change active slave
-d,--detach        Remove slave interface from bonding device
-f,--force         Force, even if interface is not Ethernet
```

ifplugd

ifplugd [OPTIONS]

Network interface plug detection daemon

Options:

```
-n      Don't daemonize
-s      Don't log to syslog
-i IFACE Interface
-f/-F   Treat link detection error as link down/link up
        (otherwise exit on error)
-a      Don't up interface at each link probe
-M      Monitor creation/destruction of interface
        (otherwise it must exist)
-r PROG Script to run
-x ARG  Extra argument for script
-I      Don't exit on nonzero exit code from script
-p      Don't run script on daemon startup
-q      Don't run script on daemon quit
-l      Run script on startup even if no cable is detected
-t SECS Poll time in seconds
-u SECS Delay before running script after link up
-d SECS Delay after link down
-m MODE API mode (mii, priv, ethtool, wlan, iff, auto)
-k      Kill running daemon
```

ifup

ifup [-anmvf] [-i FILE] IFACE...

Options:

```
-a      De/configuration all interfaces automatically
-i FILE Use FILE for interface definitions
-n      Print out what would happen, but don't do it
        (note: doesn't disable mappings)
-m      Don't run any mappings
-v      Print out what would happen before doing it
-f      Force de/configuration
```

inetd

inetd [-fe] [-q N] [-R N] [CONFFILE]

Listen for network connections and launch programs

Options:

```
-f      Run in foreground
-e      Log to stderr
-q N    Socket listen queue (default: 128)
-R N    Pause services after N connects/min
        (default: 0 - disabled)
```

init

init

Init is the parent of all processes

inotifyd

inotifyd PROG FILE1[:MASK]...

Run PROG on filesystem changes. When a filesystem event matching MASK occurs on FILEn, PROG ACTUAL_EVENTS FILEn [SUBFILE] is run. Events:

a	File is accessed
c	File is modified
e	Metadata changed
w	Writable file is closed
0	Unwritable file is closed
r	File is opened
D	File is deleted
M	File is moved
u	Backing fs is unmounted
o	Event queue overflowed
x	File can't be watched anymore

If watching a directory:

m	Subfile is moved into dir
y	Subfile is moved out of dir
n	Subfile is created
d	Subfile is deleted

inotifyd waits for PROG to exit. When x event happens for all FILES, inotifyd exits.

insmod

insmod [-qfwrsv] MODULE [symbol=value]...

Options:

-r	Remove MODULE (stacks) or do autoclean
-q	Quiet
-v	Verbose
-f	Force
-w	Wait for unload
-s	Report via syslog instead of stderr

install

install [-cdDsp] [-o USER] [-g GRP] [-m MODE] [SOURCE]... DEST

Copy files and set attributes

Options:

-c	Just copy (default)
-d	Create directories
-D	Create leading target directories
-s	Strip symbol table
-p	Preserve date
-o USER	Set ownership
-g GRP	Set group ownership
-m MODE	Set permissions

ionice

ionice [-c 1-3] [-n 0-7] [-p PID] [PROG]

Change I/O priority and class

Options:

-c	Class. 1:realtime 2:best-effort 3:idle
-n	Priority

iostat

`iostat [-c] [-d] [-t] [-z] [-k|-m] [ALL|BLOCKDEV...] [INTERVAL [COUNT]]`

Report CPU and I/O statistics

Options:

- c Show CPU utilization
- d Show device utilization
- t Print current time
- z Omit devices with no activity
- k Use kb/s
- m Use Mb/s

ip

`ip [OPTIONS] {address | route | link | tunnel | rule} {COMMAND}`

`ip [OPTIONS] OBJECT {COMMAND}` where `OBJECT := {address | route | link | tunnel | rule}` `OPTIONS := { -f[amily] { inet | inet6 | link } | -o[neline] }`

ipaddr

`ipaddr { {add|del} IFADDR dev STRING | {show|flush} [dev STRING] [to PREFIX] }`

`ipaddr {add|delete} IFADDR dev STRING ipaddr {show|flush} [dev STRING] [scope SCOPE-ID] [to PREFIX] [label PATTERN] IFADDR := PREFIX | ADDR peer PREFIX [broadcast ADDR] [anycast ADDR] [label STRING] [scope SCOPE-ID] SCOPE-ID := [host | link | global | NUMBER]`

ipcalc

`ipcalc [OPTIONS] ADDRESS[[/]NETMASK] [NETMASK]`

Calculate IP network settings from a IP address

Options:

- b,--broadcast Display calculated broadcast address
- n,--network Display calculated network address
- m,--netmask Display default netmask for IP
- p,--prefix Display the prefix for IP/NETMASK
- h,--hostname Display first resolved host name
- s,--silent Don't ever display error messages

ipcrm

`ipcrm [-MQS key] [-mq s id]`

Upper-case options MQS remove an object by shmkey value. Lower-case options remove an object by shmid value.

Options:

- mM Remove memory segment after last detach
- qQ Remove message queue
- sS Remove semaphore

ipcs

`ipcs [[-smq] -i shmid] | [[-asmq] [-tcplu]]`

-i Show specific resource

Resource specification:

- m Shared memory segments
- q Message queues
- s Semaphore arrays
- a All (default)

Output format:

- t Time
- c Creator
- p Pid

```
-l      Limits
-u      Summary
```

iplink

```
iplink { set DEVICE { up | down | arp { on | off } | show [DEVICE] }
```

```
iplink set DEVICE { up | down | arp | multicast { on | off } | dynamic { on | off } | mtu MTU } iplink show [DEVICE]
```

iproute

```
iproute { list | flush | { add | del | change | append |
                                replace | monitor } ROUTE }
```

```
iproute { list | flush } SELECTOR iproute get ADDRESS [from ADDRESS iif STRING] [oif STRING] [tos TOS]
iproute { add | del | change | append | replace | monitor } ROUTE SELECTOR := [root PREFIX] [match PREFIX]
[proto RTPROTO] ROUTE := [TYPE] PREFIX [tos TOS] [proto RTPROTO] [metric METRIC]
```

iprule

```
iprule {[list | add | del] RULE}
```

```
iprule [list | add | del] SELECTOR
ACTION
```

```
SELECTOR := [from PREFIX] [to PREFIX] [tos TOS] [fwmark
FWMARK]
[dev STRING] [pref NUMBER]
ACTION := [table TABLE_ID] [nat ADDRESS]
[prohibit | reject | unreachable]
[realms [SRCREALM/]DSTREALM]
TABLE_ID := [local | main | default | NUMBER]
```

iptunnel

```
iptunnel { add | change | del | show } [NAME]
[mode { ipip | gre | sit }]
[remote ADDR] [local ADDR] [ttl TTL]
```

```
iptunnel { add | change | del | show } [NAME]
[mode { ipip | gre | sit }] [remote ADDR] [local ADDR]
[[i|o]seq] [[i|o]key KEY] [[i|o]csum]
[ttl TTL] [tos TOS] [[no]pmtudisc] [dev PHYS_DEV]
```

kbd_mode

```
kbd_mode [-a|k|s|u] [-C TTY]
```

Report or set the keyboard mode

Options:

```
-a      Default (ASCII)
-k      Medium-raw (keyboard)
-s      Raw (scancode)
-u      Unicode (utf-8)
-C TTY  Affect TTY instead of /dev/tty
```

kill

```
kill [-l] [-SIG] PID...
```

Send a signal (default: TERM) to given PIDs

Options:

```
-l      List all signal names and numbers
```

killall

```
killall [-l] [-q] [-SIG] PROCESS_NAME...
```

Send a signal (default: TERM) to given processes

Options:

```
-l      List all signal names and numbers
-q      Don't complain if no processes were killed
```

killall5

killall5 [-l] [-SIG] [-o PID]...

Send a signal (default: TERM) to all processes outside current session

Options:

```
-l      List all signal names and numbers
-o PID  Don't signal this PID
```

klogd

klogd [-c N] [-n]

Kernel logger

Options:

```
-c N    Only messages with level < N are printed to console
-n      Run in foreground
```

last

last [-HW] [-f FILE]

Show listing of the last users that logged into the system

Options:

```
-W      Display with no host column truncation
-f FILE Read from FILE instead of /var/log/wtmp
```

length

length STRING

Print STRING's length

less

less [-EMNmh~I?] [FILE]...

View FILE (or stdin) one screenful at a time

Options:

```
-E      Quit once the end of a file is reached
-M, -m Display status line with line numbers
        and percentage through the file
-N      Prefix line number to each line
-I      Ignore case in all searches
-~      Suppress ~s displayed past the end of the file
```

ln

ln [OPTIONS] TARGET... LINK|DIR

Create a link LINK or DIR/TARGET to the specified TARGET(s)

Options:

```
-s      Make symlinks instead of hardlinks
-f      Remove existing destinations
-n      Don't dereference symlinks - treat like normal file
```

- b Make a backup of the target (if exists) before link operation
- S suf Use suffix instead of ~ when making backup files

loadfont

loadfont < font

Load a console font from stdin

loadkmap

loadkmap < keymap

Load a binary keyboard translation table from stdin

logger

logger [OPTIONS] [MESSAGE]

Write MESSAGE (or stdin) to syslog

Options:

- s Log to stderr as well as the system log
- t TAG Log using the specified tag (defaults to user name)
- p PRI0 Priority (numeric or facility.level pair)

login

login [-p] [-h HOST] [[-f] USER]

Begin a new session on the system

Options:

- f Don't authenticate (user already authenticated)
- h Name of the remote host
- p Preserve environment

logname

logname

Print the name of the current user

logread

logread [-f]

Show messages in syslogd's circular buffer

Options:

- f Output data as log grows

losetup

losetup [-o OFS] LOOPDEV FILE - associate loop devices

losetup -d LOOPDEV - disassociate

losetup [-f] - show

Options:

- o OFS Start OFS bytes into FILE
- f Show first free loop device

lpd

lpd SPOOLDIR [HELPER [ARGS]]

SPOOLDIR must contain (symlinks to) device nodes or directories with names matching print queue names. In the first case, jobs are sent directly to the device. Otherwise each job is stored in queue directory and HELPER program is

called. Name of file to print is passed in \$DATAFILE variable. Example:

```
tcpsvd -E 0 515 softlimit -m 999999 lpd /var/spool ./print
```

lpq

```
lpq [-P queue[@host[:port]]] [-U USERNAME] [-d JOBID]... [-fs]
```

Options:

```
-P      lp service to connect to (else uses $PRINTER)
-d      Delete jobs
-f      Force any waiting job to be printed
-s      Short display
```

lpr

```
lpr -P queue[@host[:port]] -U USERNAME -J TITLE -Vmh [FILE]...
```

Options:

```
-P      lp service to connect to (else uses $PRINTER)
-m      Send mail on completion
-h      Print banner page too
-V      Verbose
```

ls

```
ls [-1AacCdeFilnpLRrSsTtuvwxXhk] [FILE]...
```

List directory contents

Options:

```
-l      List in a single column
-A      Don't list . and ..
-a      Don't hide entries starting with .
-C      List by columns
-c      With -l: sort by ctime
--color[={always,never,auto}] Control coloring
-d      List directory entries instead of contents
-e      List full date and time
-F      Append indicator (one of */=@|) to entries
-i      List inode numbers
-l      Long listing format
-n      List numeric UIDs and GIDs instead of names
-p      Append indicator (one of */=@|) to entries
-L      List entries pointed to by symlinks
-R      Recurse
-r      Sort in reverse order
-S      Sort by file size
-s      List the size of each file, in blocks
-T N    Assume tabstop every N columns
-t      With -l: sort by modification time
-u      With -l: sort by access time
-v      Sort by version
-w N    Assume the terminal is N columns wide
-x      List by lines
-X      Sort by extension
-h      List sizes in human readable format (1K 243M 2G)
```

lsattr

```
lsattr [-Radlv] [FILE]...
```

List file attributes on an ext2 fs

Options:

```
-R      Recurse
-a      Don't hide entries starting with .
-d      List directory entries instead of contents
-l      List long flag names
-v      List the file's version/generation number
```

lsmod

lsmod [-qfwrsv] MODULE [symbol=value]...

Options:

-r	Remove MODULE (stacks) or do autoclean
-q	Quiet
-v	Verbose
-f	Force
-w	Wait for unload
-s	Report via syslog instead of stderr

lspci

lspci [-mk]

List all PCI devices

-m	Parseable output
-k	Show driver

lzcat

lzcat FILE

Decompress to stdout

lzma

lzma -d [-cf] [FILE]...

Decompress FILE (or stdin)

Options:

-d	Decompress
-c	Write to stdout
-f	Force

lzop

lzop [-cfvd123456789CF] [FILE]...

Options:

-1..9	Compression level
-d	Decompress
-c	Write to stdout
-f	Force
-v	Verbose
-F	Don't store or verify checksum
-C	Also write checksum of compressed block

lzopcat

lzopcat [-vCF] [FILE]...

-v	Verbose
-F	Don't store or verify checksum

makedevs

makedevs [-d device_table] rootdir

Create a range of special files as specified in a device table. Device table entries take the form of:

<type> <mode> <uid> <gid> <major> <minor> <start> <inc> <count> Where name is the file name, type can be one of: f Regular file d Directory c Character device b Block device p Fifo (named pipe) uid is the user id for the target file, gid is the group id for the target file. The rest of the entries (major, minor, etc) apply to to device special files. A '-' may be used for blank entries.

makemime

makemime [OPTIONS] [FILE]...

Create multipart MIME-encoded message from FILEs

Options:

```
-o FILE Output. Default: stdout
-a HDR  Add header. Examples:
        "From: user@host.org", "Date: `date -R`"
-c CT   Content type. Default: text/plain
-C CS   Charset. Default: us-ascii
```

Other options are silently ignored

man

man [-aw] [MANPAGE]...

Format and display manual page

Options:

```
-a      Display all pages
-w      Show page locations
```

md5sum

md5sum [FILE]... or: **md5sum** -c [-sw] [FILE]

Print or check MD5 checksums

Options:

```
-c      Check sums against given list
-s      Don't output anything, status code shows success
-w      Warn about improperly formatted checksum lines
```

mdev

mdev [-s]

```
-s      Scan /sys and populate /dev during system boot
```

It can be run by kernel as a hotplug helper. To activate it: `echo /sbin/mdev > /proc/sys/kernel/hotplug` It uses `/etc/mdev.conf` with lines `[-]DEVNAME UID:GID PERM [>|=PATH] [@$]*PROG`

mesg

mesg [y|n]

Control write access to your terminal

```
yAllow write access to your terminal
nDisallow write access to your terminal
```

microcom

microcom [-d DELAY] [-t TIMEOUT] [-s SPEED] [-X] TTY

Copy bytes for stdin to TTY and from TTY to stdout

Options:

```
-d      Wait up to DELAY ms for TTY output before sending every
        next byte to it
-t      Exit if both stdin and TTY are silent for TIMEOUT ms
-s      Set serial line to SPEED
-X      Disable special meaning of NUL and Ctrl-X from stdin
```

mkdir

mkdir [OPTIONS] DIRECTORY...

Create DIRECTORY

Options:

```
-m MODE Mode
-p      No error if exists; make parent directories as needed
```

mkdosfs

mkdosfs [-v] [-n LABEL] BLOCKDEV [KBYTES]

Make a FAT32 filesystem

Options:

```
-v      Verbose
-n LBL  Volume label
```

mke2fs

mke2fs [-Fn] [-b BLK_SIZE] [-i INODE_RATIO] [-I INODE_SIZE] [-m RESERVED_PERCENT] [-L LABEL] BLOCKDEV [KBYTES]

```
-b BLK_SIZE    Block size, bytes
-F            Force
-i RATIO      Max number of files is filesystem_size / RATIO
-I BYTES      Inode size (min 128)
-L LBL        Volume label
-m PERCENT    Percent of blocks to reserve for admin
-n           Dry run
```

mkfifo

mkfifo [-m MODE] NAME

Create named pipe

Options:

```
-m MODE Mode (default a=rw)
```

mkfs.ext2

mkfs.ext2 [-Fn] [-b BLK_SIZE] [-i INODE_RATIO] [-I INODE_SIZE] [-m RESERVED_PERCENT] [-L LABEL] BLOCKDEV [KBYTES]

```
-b BLK_SIZE    Block size, bytes
-F            Force
-i RATIO      Max number of files is filesystem_size / RATIO
-I BYTES      Inode size (min 128)
-L LBL        Volume label
-m PERCENT    Percent of blocks to reserve for admin
-n           Dry run
```

mkfs.minix

mkfs.minix [-c | -l FILE] [-nXX] [-iXX] BLOCKDEV [KBYTES]

Make a MINIX filesystem

Options:

```
-c          Check device for bad blocks
-n [14|30]  Maximum length of filenames
-i INODES   Number of inodes for the filesystem
-l FILE     Read bad blocks list from FILE
-v          Make version 2 filesystem
```

mkfs.reiser

mkfs.reiser [-f] [-l LABEL] BLOCKDEV [4K-BLOCKS]

Make a ReiserFS V3 filesystem

Options:

```
-f          Force
-l LBL     Volume label
```

mkfs.vfat

```
mkfs.vfat [-v] [-n LABEL] BLOCKDEV [KBYTES]
```

Make a FAT32 filesystem

Options:

```
-v          Verbose
-n LBL     Volume label
```

mknod

```
mknod [-m MODE] NAME TYPE MAJOR MINOR
```

Create a special file (block, character, or pipe)

Options:

```
-m MODE Creation mode (default a=rw)

TYPE:
b          Block device
c or u    Character device
p          Named pipe (MAJOR and MINOR are ignored)
```

mkpasswd

```
mkpasswd [OPTIONS] [PASSWORD] [SALT]
```

Crypt the PASSWORD using crypt(3)

Options:

```
-P, --password-fd=N    Read password from fd N
-m, --method=TYPE      Encryption method TYPE
-S, --salt=SALT
```

mkswap

```
mkswap [-L LBL] BLOCKDEV [KBYTES]
```

Prepare BLOCKDEV to be used as swap partition

Options:

```
-L LBL  Label
```

mktemp

```
mktemp [-dt] [-p DIR] [TEMPLATE]
```

Create a temporary file with name based on TEMPLATE and print its name. TEMPLATE must end with XXXXXX (e.g. [/dir/]nameXXXXXX).

Options:

```
-d          Make a directory instead of a file
-t          Generate a path rooted in temporary directory
-p DIR      Use DIR as a temporary directory (implies -t)
```

For -t or -p, directory is chosen as follows: \$TMPDIR if set, else -p DIR, else /tmp

modinfo

```
modinfo [-adlp0] [-F keyword] MODULE
```

Options:

```

-a          Shortcut for '-F author'
-d          Shortcut for '-F description'
-l          Shortcut for '-F license'
-p          Shortcut for '-F parm'
-F keyword  Keyword to look for
-0          Separate output with NULs

```

modprobe

modprobe [-qfwrsv] MODULE [symbol=value]...

Options:

```

-r          Remove MODULE (stacks) or do autoclean
-q          Quiet
-v          Verbose
-f          Force
-w          Wait for unload
-s          Report via syslog instead of stderr

```

more

more [FILE]...

View FILE (or stdin) one screenful at a time

mount

mount [OPTIONS] [-o OPTS] DEVICE NODE

Mount a filesystem. Filesystem autodetection requires /proc.

Options:

```

-a          Mount all filesystems in fstab
-f          Update /etc/mtab, but don't mount
-i          Don't run mount helper
-n          Don't update /etc/mtab
-r          Read-only mount
-w          Read-write mount (default)
-t FSTYPE   Filesystem type
-O OPT      Mount only filesystems with option OPT (-a only)
-o OPT:
loop        Ignored (loop devices are autodetected)
[a]sync     Writes are [a]synchronous
[no]atime   Disable/enable updates to inode access times
[no]diratime Disable/enable atime updates to directories
[no]relatime Disable/enable atime updates relative to modification time
[no]dev     (Dis)allow use of special device files
[no]exec    (Dis)allow use of executable files
[no]suid    (Dis)allow set-user-id-root programs
[r]shared   Convert [recursively] to a shared subtree
[r]slave    Convert [recursively] to a slave subtree
[r]private  Convert [recursively] to a private subtree
[un]bindable Make mount point [un]able to be bind mounted
bind        Bind a file or directory to another location
move        Relocate an existing mount point
remount     Remount a mounted filesystem, changing flags
ro/rw       Same as -r/-w

```

There are filesystem-specific -o flags.

mountpoint

mountpoint [-q] <[-dn] DIR | -x DEVICE>

Check if the directory is a mountpoint

Options:

```

-q          Quiet
-d          Print major/minor device number of the filesystem
-n          Print device name of the filesystem
-x          Print major/minor device number of the blockdevice

```

mpstat

mpstat [-A] [-I SUM|CPU|ALL|SCPU] [-u] [-P num|ALL] [INTERVAL [COUNT]]

Per-processor statistics

Options:

-A	Same as -I ALL -u -P ALL
-I SUM CPU ALL SCPU	Report interrupt statistics
-P num ALL	Processor to monitor
-u	Report CPU utilization

mt

mt [-f device] opcode value

Control magnetic tape drive operation

Available Opcodes:

bsf bsfm bsr bss datacompression drvbuffer eof eom erase fsf fsfm fsr fss load lock mkpart nop offline ras1 ras2 ras3
reset retension rewind rewoffline seek setblk setdensity setpart tell unload unlock weof wset

mv

mv [-fin] SOURCE DEST or: mv [-fin] SOURCE... DIRECTORY

Rename SOURCE to DEST, or move SOURCE(s) to DIRECTORY

Options:

-f	Don't prompt before overwriting
-i	Interactive, prompt before overwrite
-n	Don't overwrite an existing file

nameif

nameif [-s] [-c FILE] [{IFNAME MACADDR}]

Rename network interface while it in the down state

Options:

-c FILE	Use configuration file (default: /etc/mactab)
-s	Use syslog (LOCAL0 facility)
IFNAME MACADDR	new_interface_name interface_mac_address

nanddump

nanddump [-o] [-b] [-s ADDR] [-f FILE] MTD_DEVICE

Dump the sepcified MTD device

Options:

-o	Omit oob data
-b	Omit bad block from the dump
-s ADDR	Start address
-l LEN	Length
-f FILE	Dump to file ('-' for stdout)

nandwrite

nandwrite [-p] [-s ADDR] MTD_DEVICE [FILE]

Write to the specified MTD device

Options:

-p	Pad to page size
-s ADDR	Start address

nbd-client

nbd-client HOST PORT BLOCKDEV

Connect to HOST and provide a network block device on BLOCKDEV

nc

nc [OPTIONS] HOST PORT - connect nc [OPTIONS] -l -p PORT [HOST] [PORT] - listen

Options:

- e PROG Run PROG after connect (must be last)
- l Listen mode, for inbound connects
- p PORT Local port
- s ADDR Local address
- w SEC Timeout for connects and final net reads
- i SEC Delay interval for lines sent
- n Don't do DNS resolution
- u UDP mode
- v Verbose
- o FILE Hex dump traffic
- z Zero-I/O mode (scanning)

netstat

netstat [-ral] [-tuwx] [-enWp]

Display networking information

Options:

- r Routing table
- a All sockets
- l Listening sockets
- Else: connected sockets
- t TCP sockets
- u UDP sockets
- w Raw sockets
- x Unix sockets
- Else: all socket types
- e Other/more information
- n Don't resolve names
- W Wide display
- p Show PID/program name for sockets

nice

nice [-n ADJUST] [PROG ARGS]

Change scheduling priority, run PROG

Options:

- n ADJUST Adjust priority by ADJUST

nmeter

nmeter format_string

Monitor system in real time

Format specifiers:

- %Nc or %[cN] Monitor CPU. N - bar size, default 10
(displays: S:system U:user N:niced D:iowait I:irq i:softirq)
- %[niface] Monitor network interface 'iface'
- %m Monitor allocated memory
- %[mf] Monitor free memory
- %[mt] Monitor total memory
- %s Monitor allocated swap
- %f Monitor number of used file descriptors
- %Ni Monitor total/specific IRQ rate
- %x Monitor context switch rate

```
%p      Monitor forks
%[pn]   Monitor # of processes
%b      Monitor block io
%Nt     Show time (with N decimal points)
%Nd     Milliseconds between updates (default:1000)
%r      Print <cr> instead of <lf> at EOL
```

nohup

nohup PROG ARGS

Run PROG immune to hangups, with output to a non-tty

nslookup

nslookup [HOST] [SERVER]

Query the nameserver for the IP address of the given HOST optionally using a specified DNS server

ntpd

ntpd [-dnqNwl] [-S PROG] [-p PEER]...

NTP client/server

Options:

```
-d      Verbose
-n      Do not daemonize
-q      Quit after clock is set
-N      Run at high priority
-w      Do not set time (only query peers), implies -n
-l      Run as server on port 123
-S PROG Run PROG after stepping time, stratum change, and every 11 mins
-p PEER Obtain time from PEER (may be repeated)
```

od

od [-aBbcDdeFfHhIiLlOovXx] [-t TYPE] [FILE]

Write an unambiguous representation, octal bytes by default, of FILE (or stdin) to stdout

openvt

openvt [-c N] [-sw] [PROG ARGS]

Start PROG on a new virtual terminal

Options:

```
-c N    Use specified VT
-s      Switch to the VT
-w      Wait for PROG to exit
```

passwd

passwd [OPTIONS] [USER]

Change USER's password. If no USER is specified, changes the password for the current user.

Options:

```
-a ALG  Algorithm to use for password (des, md5)
-d      Delete password for the account
-l      Lock (disable) account
-u      Unlock (re-enable) account
```

patch

patch [OPTIONS] [ORIGFILE [PATCHFILE]]

```
-p,--strip N      Strip N leading components from file names
-i,--input DIFF   Read DIFF instead of stdin
```

-R,--reverse	Reverse patch
-N,--forward	Ignore already applied patches
--dry-run	Don't actually change files
-E,--remove-empty-files	Remove output files if they become empty

pgrep

pgrep [-flnovx] [-s SID|-P PPID]PATTERN]

Display process(es) selected by regex PATTERN

Options:

-l	Show command name too
-f	Match against entire command line
-n	Show the newest process only
-o	Show the oldest process only
-v	Negate the match
-x	Match whole name (not substring)
-s	Match session ID (0 for current)
-P	Match parent process ID

pidof

pidof [OPTIONS] [NAME]...

List PIDs of all processes with names that match NAMES

Options:

-s	Show only one PID
-o PID	Omit given pid
	Use %PPID to omit pid of pidof's parent

ping

ping [OPTIONS] HOST

Send ICMP ECHO_REQUEST packets to network hosts

Options:

-4, -6	Force IP or IPv6 name resolution
-c CNT	Send only CNT pings
-s SIZE	Send SIZE data bytes in packets (default:56)
-I IFACE/IP	Use interface or IP address as source
-W SEC	Seconds to wait for the first response (default:10) (after all -c CNT packets are sent)
-w SEC	Seconds until ping exits (default:infinite) (can exit earlier with -c CNT)
-q	Quiet, only displays output at start and when finished

ping6

ping6 [OPTIONS] HOST

Send ICMP ECHO_REQUEST packets to network hosts

Options:

-c CNT	Send only CNT pings
-s SIZE	Send SIZE data bytes in packets (default:56)
-I IFACE/IP	Use interface or IP address as source
-q	Quiet, only displays output at start and when finished

pivot_root

pivot_root NEW_ROOT PUT_OLD

Move the current root file system to PUT_OLD and make NEW_ROOT the new root file system

pkill

kill [-l|-SIGNAL] [-fnovx] [-s SID|-P PPID|PATTERN]

Send a signal to process(es) selected by regex PATTERN

Options:

- l List all signals
- f Match against entire command line
- n Signal the newest process only
- o Signal the oldest process only
- v Negate the match
- x Match whole name (not substring)
- s Match session ID (0 for current)
- P Match parent process ID

pmap

pmap [-x][[-q] PID

Display detailed precesses' memory usage

Options:

- x show details
- q quiet

popmaildir

popmaildir [OPTIONS] MAILDIR [CONN_HELPER ARGS]

Fetch content of remote mailbox to local maildir

Options:

- s Skip authorization
- T Get messages with TOP instead of RETR
- k Keep retrieved messages on the server
- t SEC Network timeout
- F "PROG ARGS" Filter program (may be repeated)
- M "PROG ARGS" Delivery program

Fetch from plain POP3 server: **popmaildir -k DIR nc pop3.server.com 110 <user_and_pass.txt** Fetch from SSLeD POP3 server and delete fetched emails: **popmaildir DIR -- openssl s_client -quiet -connect pop3.server.com:995 <user_and_pass.txt**

poweroff

poweroff [-d DELAY] [-n] [-f]

Halt and shut off power

Options:

- d SEC Delay interval
- n Do not sync
- f Force (don't go through init)

powertop

powertop

Analyze power consumption on Intel-based laptops

printenv

printenv [VARIABLE]...

Print environment VARIABLES. If no VARIABLE specified, print all.

printf

printf FORMAT [ARGUMENT]...

Format and print ARGUMENT(s) according to FORMAT, where FORMAT controls the output exactly as in C printf

ps

ps [-o COL1,COL2=HEADER] [-T]

Show list of processes

Options:

-o COL1,COL2=HEADER	Select columns for display
-T	Show threads

pscan

pscan [-cb] [-p MIN_PORT] [-P MAX_PORT] [-t TIMEOUT] [-T MIN_RTT] HOST

Scan a host, print all open ports

Options:

-c	Show closed ports too
-b	Show blocked ports too
-p	Scan from this port (default 1)
-P	Scan up to this port (default 1024)
-t	Timeout (default 5000 ms)
-T	Minimum rtt (default 5 ms, increase for congested hosts)

pwd

pwd

Print the full filename of the current working directory

raidautorun

raidautorun DEVICE

Tell the kernel to automatically search and start RAID arrays

rdate

rdate [-sp] HOST

Get and possibly set the system date and time from a remote HOST

Options:

-s	Set the system date and time (default)
-p	Print the date and time

rdev

rdev

Print the device node associated with the filesystem mounted at '/'

readahead

readahead [FILE]...

Preload FILEs to RAM

readlink

readlink [-fnv] FILE

Display the value of a symlink

Options:

```
-f Canonicalize by following all symlinks
-n Don't add newline
-v Verbose
```

readprofile

readprofile [OPTIONS]

Options:

```
-m mapfile      (Default: /boot/System.map)
-p profile      (Default: /proc/profile)
-M NUM          Set the profiling multiplier to NUM
-i             Print only info about the sampling step
-v             Verbose
-a             Print all symbols, even if count is 0
-b             Print individual histogram-bin counts
-s             Print individual counters within functions
-r             Reset all the counters (root only)
-n             Disable byte order auto-detection
```

realpath

realpath FILE...

Return the absolute pathnames of given FILE

reboot

reboot [-d DELAY] [-n] [-f]

Reboot the system

Options:

```
-d SEC Delay interval
-n Do not sync
-f Force (don't go through init)
```

reformime

reformime [OPTIONS] [FILE]...

Parse MIME-encoded message

Options:

```
-x PREFIX      Extract content of MIME sections to files
-X PROG ARGS   Filter content of MIME sections through PROG
               Must be the last option
```

Other options are silently ignored

remove-shell

remove-shell SHELL...

Remove SHELLs from /etc/shells

renice

renice {{-n INCREMENT} | PRIORITY} [[-p | -g | -u] ID...]

Change scheduling priority for a running process

Options:

```
-n Adjust current nice value (smaller is faster)
-p Process id(s) (default)
-g Process group id(s)
-u Process user name(s) and/or id(s)
```

reset

reset

Reset the screen

resize

resize

Resize the screen

rev

rev [FILE]...

Reverse lines of FILE

rfkill

rfkill COMMAND [INDEX|TYPE]

Enable/disable wireless devices

Commands:

list [INDEX TYPE]	List current state
block INDEX TYPE	Disable device
unblock INDEX TYPE	Enable device

TYPE: all, wlan(wifi), bluetooth, uwb(ultrawideband),
wimax, wwan, gps, fm

rm

rm [-irf] FILE...

Remove (unlink) FILEs

Options:

-i	Always prompt before removing
-f	Never prompt
-R, -r	Recurse

rmdir

rmdir [OPTIONS] DIRECTORY...

Remove DIRECTORY if it is empty

Options:

-p --parents	Include parents
--ignore-fail-on-non-empty	

rmmod

rmmod [-qfwrsv] MODULE [symbol=value]...

Options:

-r	Remove MODULE (stacks) or do autoclean
-q	Quiet
-v	Verbose
-f	Force
-w	Wait for unload
-s	Report via syslog instead of stderr

route

route [{add|del|delete}]

Edit kernel routing tables

Options:

```
-n      Don't resolve names
-e      Display other/more information
-A inet{6}  Select address family
```

rpm

rpm -i PACKAGE.rpm; rpm -qp[i ldc] PACKAGE.rpm

Manipulate RPM packages

Commands:

```
-i      Install package
-qp     Query package

Options:

-i      Show information
-l      List contents
-d      List documents
-c      List config files
```

rpm2cpio

rpm2cpio package.rpm

Output a cpio archive of the rpm file

rtcwake

rtcwake [-a | -l | -u] [-d DEV] [-m MODE] [-s SEC | -t TIME]

Enter a system sleep state until specified wakeup time

```
-a,--auto      Read clock mode from adjtime
-l,--local     Clock is set to local time
-u,--utc       Clock is set to UTC time
-d,--device=DEV Specify the RTC device
-m,--mode=MODE Set the sleep state (default: standby)
-s,--seconds=SEC Set the timeout in SEC seconds from now
-t,--time=TIME Set the timeout to TIME seconds from epoch
```

run-parts

run-parts [-t] [-l] [-a ARG] [-u MASK] DIRECTORY

Run a bunch of scripts in DIRECTORY

Options:

```
-t      Print what would be run, but don't actually run anything
-a ARG  Pass ARG as argument for every program
-u MASK Set the umask to MASK before running every program
-l      Print names of all matching files even if they are not executable
```

runlevel

runlevel [FILE]

Find the current and previous system runlevel

If no utmp FILE exists or if no runlevel record can be found, print "unknown"

runsv

runsv DIR

Start and monitor a service and optionally an appendant log service

runsvdir

runsvdir [-P] [-s SCRIPT] DIR

Start a runsv process for each subdirectory. If it exits, restart it.

```
-P          Put each runsv in a new session
-s SCRIPT  Run SCRIPT <signo> after signal is processed
```

rx

rx FILE

Receive a file using the xmodem protocol

script

script [-afqt] [-c PROG] [OUTFILE]

Options:

```
-a          Append output
-c PROG    Run PROG, not shell
-f          Flush output after each write
-q          Quiet
-t          Send timing to stderr
```

scriptreplay

scriptreplay timingfile [typescript [divisor]]

Play back typescripts, using timing information

sed

sed [-efinr] SED_CMD [FILE]...

Options:

```
-e CMD     Add CMD to sed commands to be executed
-f FILE    Add FILE contents to sed commands to be executed
-i          Edit files in-place (else sends result to stdout)
-n          Suppress automatic printing of pattern space
-r          Use extended regex syntax
```

If no -e or -f, the first non-option argument is the sed command string. Remaining arguments are input files (stdin if none).

sendmail

sendmail [OPTIONS] [RECIPIENT_EMAIL]...

Read email from stdin and send it

Standard options:

```
-t          Read additional recipients from message body
-f sender   Sender (required)
-o options  Various options. -oi implied, others are ignored
-i          -oi synonym. implied and ignored
```

Busybox specific options:

```
-w seconds  Network timeout
-H 'PROG ARGS' Run connection helper
Examples:
-H 'exec openssl s_client -quiet -tls1 -starttls smtp
    -connect smtp.gmail.com:25' <email.txt
    [4<username_and_passwd.txt | -au<username> -ap<password>]
-H 'exec openssl s_client -quiet -tls1
    -connect smtp.gmail.com:465' <email.txt
    [4<username_and_passwd.txt | -au<username> -ap<password>]
-S server[:port] Server
-au<username> Username for AUTH LOGIN
-ap<password> Password for AUTH LOGIN
-am<method>  Authentication method. Ignored. LOGIN is implied
```

Other options are silently ignored; -oi -t is implied Use makemime applet to create message with attachments

seq

seq [-w] [-s SEP] [FIRST [INC]] LAST

Print numbers from FIRST to LAST, in steps of INC. FIRST, INC default to 1.

Options:

-w	Pad to last with leading zeros
-s SEP	String separator

setarch

setarch personality PROG ARGS

Personality may be:

linux32	Set 32bit uname emulation
linux64	Set 64bit uname emulation

setconsole

setconsole [-r|--reset] [DEVICE]

Redirect system console output to DEVICE (default: /dev/tty)

Options:

-r	Reset output to /dev/console
----	------------------------------

setfont

setfont FONT [-m MAPFILE] [-C TTY]

Load a console font

Options:

-m MAPFILE	Load console screen map
-C TTY	Affect TTY instead of /dev/tty

setkeycodes

setkeycodes SCANCODE KEYCODE...

Set entries into the kernel's scancode-to-keycode map, allowing unusual keyboards to generate usable keycodes.

SCANCODE may be either xx or e0xx (hexadecimal), and KEYCODE is given in decimal.

setlogcons

setlogcons N

Redirect the kernel output to console N (0 for current)

setsid

setsid PROG ARGS

Run PROG in a new session. PROG will have no controlling terminal and will not be affected by keyboard signals (Ctrl-C etc). See [setsid\(2\)](#) for details.

setuidgid

setuidgid USER PROG ARGS

Set uid and gid to USER's uid and gid, drop supplementary group ids, run PROG

sha1sum

sha1sum [FILE]... or: sha1sum -c [-sw] [FILE]

Print or check SHA1 checksums

Options:

- c Check sums against given list
- s Don't output anything, status code shows success
- w Warn about improperly formatted checksum lines

sha256sum

sha256sum [FILE]... or: sha256sum -c [-sw] [FILE]

Print or check SHA256 checksums

Options:

- c Check sums against given list
- s Don't output anything, status code shows success
- w Warn about improperly formatted checksum lines

sha512sum

sha512sum [FILE]... or: sha512sum -c [-sw] [FILE]

Print or check SHA512 checksums

Options:

- c Check sums against given list
- s Don't output anything, status code shows success
- w Warn about improperly formatted checksum lines

showkey

showkey [-a | -k | -s]

Show keys pressed

Options:

- a Display decimal/octal/hex values of the keys
- k Display interpreted keycodes (default)
- s Display raw scan-codes

slattach

slattach [-cehmLF] [-s SPEED] [-p PROTOCOL] DEVICE

Attach network interface(s) to serial line(s)

Options:

- p PROT Set protocol (slip, cslip, slip6, clisp6 or adaptive)
- s SPD Set line speed
- e Exit after initializing device
- h Exit when the carrier is lost
- c PROG Run PROG when the line is hung up
- m Do NOT initialize the line in raw 8 bits mode
- L Enable 3-wire operation
- F Disable RTS/CTS flow control

sleep

sleep [N]...

Pause for a time equal to the total of the args given, where each arg can have an optional suffix of (s)econds, (m)inutes, (h)ours, or (d)ays

smemcap

smemcap >SMEMDATA.TAR

Collect memory usage data in /proc and write it to stdout

softlimit

softlimit [-a BYTES] [-m BYTES] [-d BYTES] [-s BYTES] [-l BYTES]

[-f BYTES] [-c BYTES] [-r BYTES] [-o N] [-p N]
[-t N]
PROG ARGS

Set soft resource limits, then run PROG

Options:

-a BYTES Limit total size of all segments
-m BYTES Same as -d BYTES -s BYTES -l BYTES -a BYTES
-d BYTES Limit data segment
-s BYTES Limit stack segment
-l BYTES Limit locked memory size
-o N Limit number of open files per process
-p N Limit number of processes per uid

Options controlling file sizes:

-f BYTES Limit output file sizes
-c BYTES Limit core file size

Efficiency opts:

-r BYTES Limit resident set size
-t N Limit CPU time, process receives
a SIGXCPU after N seconds

sort

sort [-nrugMcszdbfimSTokt] [-o FILE] [-k start[.offset][opts][,end[.offset][opts]] [-t CHAR] [FILE]...

Sort lines of text

Options:

-b Ignore leading blanks
-c Check whether input is sorted
-d Dictionary order (blank or alphanumeric only)
-f Ignore case
-g General numerical sort
-i Ignore unprintable characters
-k Sort key
-M Sort month
-n Sort numbers
-o Output to file
-k Sort by key
-t CHAR Key separator
-r Reverse sort order
-s Stable (don't sort ties alphabetically)
-u Suppress duplicate lines
-z Lines are terminated by NUL, not newline
-mST Ignored for GNU compatibility

split

split [OPTIONS] [INPUT [PREFIX]]

Options:

-b N[k|m] Split by N (kilo|mega)bytes
-l N Split by N lines
-a N Use N letters as suffix

start-stop-daemon

start-stop-daemon [OPTIONS] [-S|-K] ... [-- ARGS...]

Search for matching processes, and then -K: stop all matching processes. -S: start a process unless a matching process is found.

Process matching:

```

-u,--user USERNAME|UID  Match only this user's processes
-n,--name NAME          Match processes with NAME
                        in comm field in /proc/PID/stat
-x,--exec EXECUTABLE    Match processes with this command
                        in /proc/PID/cmdline
-p,--pidfile FILE       Match a process with PID from the file
All specified conditions must match

-S only:
-x,--exec EXECUTABLE    Program to run
-a,--startas NAME       Zeroth argument
-b,--background        Background
-N,--nicelevel N        Change nice level
-c,--chuid USER[:GRP]  Change to user/group
-m,--make-pidfile       Write PID to the pidfile specified by -p

-K only:
-s,--signal SIG         Signal to send
-t,--test               Match only, exit with 0 if a process is found

Other:
-o,--oknodo             Exit with status 0 if nothing is done
-v,--verbose            Verbose
-q,--quiet              Quiet

```

stat

stat [OPTIONS] FILE...

Display file (default) or filesystem status

Options:

```

-c fmt  Use the specified format
-f      Display filesystem status
-L      Follow links
-t      Display info in terse form

```

Valid format sequences for files:

```

%a  Access rights in octal
%A  Access rights in human readable form
%b  Number of blocks allocated (see %B)
%B  The size in bytes of each block reported by %b
%d  Device number in decimal
%D  Device number in hex
%f  Raw mode in hex
%F  File type
%g  Group ID of owner
%G  Group name of owner
%h  Number of hard links
%i  Inode number
%n  File name
%N  File name, with -> TARGET if symlink
%o  I/O block size
%s  Total size, in bytes
%t  Major device type in hex
%T  Minor device type in hex
%u  User ID of owner
%U  User name of owner
%x  Time of last access
%X  Time of last access as seconds since Epoch
%y  Time of last modification
%Y  Time of last modification as seconds since Epoch
%z  Time of last change
%Z  Time of last change as seconds since Epoch

```

Valid format sequences for file systems:

```

%a  Free blocks available to non-superuser
%b  Total data blocks in file system
%c  Total file nodes in file system
%d  Free file nodes in file system
%f  Free blocks in file system
%i  File System ID in hex
%l  Maximum length of filenames
%n  File name

```

```
%s    Block size (for faster transfer)
%S    Fundamental block size (for block counts)
%t    Type in hex
%T    Type in human readable form
```

strings

strings [-afo] [-n LEN] [FILE]...

Display printable strings in a binary file

Options:

```
-a      Scan whole file (default)
-f      Precede strings with filenames
-n LEN  At least LEN characters form a string (default 4)
-o      Precede strings with decimal offsets
```

stty

stty [-a|g] [-F DEVICE] [SETTING]...

Without arguments, prints baud rate, line discipline, and deviations from stty sane

Options:

```
-F DEVICE  Open device instead of stdin
-a         Print all current settings in human-readable form
-g         Print in stty-readable form
[SETTING] See manpage
```

su

su [OPTIONS] [-] [USERNAME]

Change user id or become root

Options:

```
-p, -m  Preserve environment
-c CMD  Command to pass to 'sh -c'
-s SH   Shell to use instead of default shell
```

sulogin

sulogin [-t N] [TTY]

Single user login

Options:

```
-t N      Timeout
```

sum

sum [-rs] [FILE]...

Checksum and count the blocks in a file

Options:

```
-r      Use BSD sum algorithm (1K blocks)
-s      Use System V sum algorithm (512byte blocks)
```

sv

sv [-v] [-w SEC] CMD SERVICE_DIR...

Control services monitored by runsv supervisor. Commands (only first character is enough):

status: query service status up: if service isn't running, start it. If service stops, restart it once: like 'up', but if service stops, don't restart it down: send TERM and CONT signals. If ./run exits, start ./finish if it exists. After it stops, don't restart service exit: send TERM and CONT signals to service and log service. If they exit, runsv exits too pause, cont,

hup, alarm, interrupt, quit, 1, 2, term, kill: send STOP, CONT, HUP, ALRM, INT, QUIT, USR1, USR2, TERM, KILL signal to service

svlogd

svlogd [-ttv] [-r C] [-R CHARS] [-l MATCHLEN] [-b BUFLen] DIR...

Continuously read log data from stdin, optionally filter log messages, and write the data to one or more automatically rotated logs

swapoff

swapoff [-a] [DEVICE]

Stop swapping on DEVICE

Options:

-a Stop swapping on all swap devices

swapon

swapon [-a] [-p PRI] [DEVICE]

Start swapping on DEVICE

Options:

-a Start swapping on all swap devices

-p PRI Set swap device priority

switch_root

switch_root [-c /dev/console] NEW_ROOT NEW_INIT [ARGS]

Free initramfs and switch to another root fs:

chroot to NEW_ROOT, delete all in /, move NEW_ROOT to /, execute NEW_INIT. PID must be 1. NEW_ROOT must be a mountpoint.

Options:

-c DEV Reopen stdio to DEV after switch

sync

sync

Write all buffered blocks to disk

sysctl

sysctl [OPTIONS] [VALUE]...

Configure kernel parameters at runtime

Options:

-n Don't print key names

-e Don't warn about unknown keys

-w Change sysctl setting

-p FILE Load sysctl settings from FILE (default /etc/sysctl.conf)

-a Display all values

-A Display all values in table form

syslogd

syslogd [OPTIONS]

System logging utility. This version of syslogd ignores /etc/syslog.conf

Options:

```
-n          Run in foreground
-O FILE     Log to given file (default:/var/log/messages)
-l N        Set local log level
-S          Smaller logging output
-s SIZE     Max size (KB) before rotate (default:200KB, 0=off)
-b N        N rotated logs to keep (default:1, max=99, 0=purge)
-R HOST[:PORT] Log to IP or hostname on PORT (default PORT=514/UDP)
-L          Log locally and via network (default is network only if -R)
-D          Drop duplicates
-C[size(KiB)] Log to shared mem buffer (read it using logread)
```

tac

tac [FILE]...

Concatenate FILEs and print them in reverse

tail

tail [OPTIONS] [FILE]...

Print last 10 lines of each FILE (or stdin) to stdout. With more than one FILE, precede each with a filename header.

Options:

```
-f          Print data as file grows
-s SECONDS  Wait SECONDS between reads with -f
-n N[kbm]   Print last N lines
-c N[kbm]   Print last N bytes
-q          Never print headers
-v          Always print headers
```

N may be suffixed by k (x1024), b (x512), or m (x1024^2). If N starts with a '+', output begins with the Nth item from the start of each file, not from the end.

tar

tar [-cxtzjaZmvO] [-X FILE] [-f TARFILE] [-C DIR] [FILE]...

Create, extract, or list files from a tar file

Operation:

```
c          Create
x          Extract
t          List
```

Options:

```
f          Name of TARFILE ('-' for stdin/out)
C          Change to DIR before operation
v          Verbose
z          (De)compress using gzip
j          (De)compress using bzip2
a          (De)compress using lzma
Z          (De)compress using compress
O          Extract to stdout
h          Follow symlinks
m          Don't restore mtime
exclude    File to exclude
X          File with names to exclude
T          File with names to include
```

taskset

taskset [-p] [MASK] [PID | PROG ARGS]

Set or get CPU affinity

Options:

```
-p          Operate on an existing PID
```

tcpsvd

`tcpsvd [-hEv] [-c N] [-C N[:MSG]] [-b N] [-u USER] [-l NAME] IP PORT PROG`

Create TCP socket, bind to IP:PORT and listen for incoming connection. Run PROG for each connection.

IP	IP to listen on. '0' = all
PORT	Port to listen on
PROG ARGS	Program to run
-l NAME	Local hostname (else looks up local hostname in DNS)
-u USER[:GRP]	Change to user/group after bind
-c N	Handle up to N connections simultaneously
-b N	Allow a backlog of approximately N TCP SYNs
-C N[:MSG]	Allow only up to N connections from the same IP. New connections from this IP address are closed immediately. MSG is written to the peer before close
-h	Look up peer's hostname
-E	Don't set up environment variables
-v	Verbose

tee

`tee [-ai] [FILE]...`

Copy stdin to each FILE, and also to stdout

Options:

-a	Append to the given FILEs, don't overwrite
-i	Ignore interrupt signals (SIGINT)

telnet

`telnet [-a] [-l USER] HOST [PORT]`

Connect to telnet server

Options:

-a	Automatic login with \$USER variable
-l USER	Automatic login as USER

telnetd

`telnetd [OPTIONS]`

Handle incoming telnet connections

Options:

-l LOGIN	Exec LOGIN on connect
-f ISSUE_FILE	Display ISSUE_FILE instead of /etc/issue
-K	Close connection as soon as login exits (normally wait until all programs close slave pty)
-p PORT	Port to listen on
-b ADDR[:PORT]	Address to bind to
-F	Run in foreground
-i	Inetd mode
-w SEC	Inetd 'wait' mode, linger time SEC
-S	Log to syslog (implied by -i or without -F and -w)

test

`test EXPRESSION ]`

Check file types, compare values etc. Return a 0/1 exit code depending on logical value of EXPRESSION

tftp

`tftp [OPTIONS] HOST [PORT]`

Transfer a file from/to tftp server

Options:

```
-l FILE Local FILE
-r FILE Remote FILE
-g      Get file
-p      Put file
-b SIZE Transfer blocks of SIZE octets
```

tftpd

tftpd [-cr] [-u USER] [DIR]

Transfer a file on tftp client's request

tftpd should be used as an inetd service. tftpd's line for inetd.conf: 69 dgram udp nowait root tftpd tftpd /files/to/serve
It also can be ran from udpsvd:

```
udpsvd -vE 0.0.0.0 69 tftpd /files/to/serve
```

Options:

```
-r      Prohibit upload
-c      Allow file creation via upload
-u      Access files as USER
```

time

time [-v] PROG ARGS

Run PROG, display resource usage when it exits

Options:

```
-v      Verbose
```

timeout

timeout [-t SECS] [-s SIG] PROG ARGS

Runs PROG. Sends SIG to it if it is not gone in SECS seconds. Defaults: SECS: 10, SIG: TERM.

top

top [-b] [-nCOUNT] [-dSECONDS] [-m]

Provide a view of process activity in real time. Read the status of all processes from /proc each SECONDS and display a screenful of them.

touch

touch [-c] [-d DATE] [-r FILE] FILE [FILE]...

Update the last-modified date on the given FILE[s]

Options:

```
-c      Don't create files
-d DT   Date/time to use
-r FILE Use FILE's date/time
```

tr

tr [-cds] STRING1 [STRING2]

Translate, squeeze, or delete characters from stdin, writing to stdout

Options:

```
-c      Take complement of STRING1
-d      Delete input characters coded STRING1
-s      Squeeze multiple output characters of STRING2 into one character
```

traceroute

```
traceroute [-46fildnr] [-f 1ST_TTL] [-m MAXTTL] [-p
PORT] [-q PROBES]
```

```
[-s SRC_IP] [-t TOS] [-w WAIT_SEC] [-g
GATEWAY] [-i IFACE]
[-z PAUSE_MSEC] HOST [BYTES]
```

Trace the route to HOST

Options:

```
-4, -6    Force IP or IPv6 name resolution
-F        Set the don't fragment bit
-I        Use ICMP ECHO instead of UDP datagrams
-l        Display the TTL value of the returned packet
-d        Set SO_DEBUG options to socket
-n        Print numeric addresses
-r        Bypass routing tables, send directly to HOST
-v        Verbose
-m        Max time-to-live (max number of hops)
-p        Base UDP port number used in probes
          (default 33434)
-q        Number of probes per TTL (default 3)
-s        IP address to use as the source address
-t        Type-of-service in probe packets (default 0)
-w        Time in seconds to wait for a response (default 3)
-g        Loose source route gateway (8 max)
```

traceroute6

```
traceroute6 [-dnrv] [-m MAXTTL] [-p PORT] [-q PROBES]
```

```
[-s SRC_IP] [-t TOS] [-w WAIT_SEC] [-i IFACE]
HOST [BYTES]
```

Trace the route to HOST

Options:

```
-d        Set SO_DEBUG options to socket
-n        Print numeric addresses
-r        Bypass routing tables, send directly to HOST
-v        Verbose
-m        Max time-to-live (max number of hops)
-p        Base UDP port number used in probes
          (default is 33434)
-q        Number of probes per TTL (default 3)
-s        IP address to use as the source address
-t        Type-of-service in probe packets (default 0)
-w        Time in seconds to wait for a response (default 3)
```

true

```
true
```

Return an exit code of TRUE (0)

tty

```
tty
```

Print file name of stdin's terminal

Options:

```
-s        Print nothing, only return exit status
```

ttysize

```
ttysize [w] [h]
```

Print dimension(s) of stdin's terminal, on error return 80x25

tunctl

tunctl [-f device] ([-t name] | -d name) [-u owner] [-g group] [-b]

Create or delete tun interfaces

Options:

-f name	tun device (/dev/net/tun)
-t name	Create iface 'name'
-d name	Delete iface 'name'
-u owner	Set iface owner
-g group	Set iface group
-b	Brief output

tune2fs

tune2fs [-c MOUNT_CNT] [-i DAYS] [-L LABEL] BLOCKDEV

Adjust filesystem options on ext[23] filesystems

ubiattach

ubiattach -m MTD_NUM [-d UBI_NUM] UBI_CTRL_DEV

Attach MTD device to UBI

Options:

-m MTD_NUM	MTD device number to attach
-d UBI_NUM	UBI device number to assign

ubidetach

ubidetach -d UBI_NUM UBI_CTRL_DEV

Detach MTD device from UBI

Options:

-d UBI_NUM	UBI device number
------------	-------------------

udhcpc

udhcpc [-fbnqvoCR] [-i IFACE] [-r IP] [-s PROG] [-p PIDFILE]

[-H HOSTNAME] [-V VENDOR] [-x OPT:VAL]... [-O OPT]... [-P N]

-i,--interface IFACE	Interface to use (default eth0)
-p,--pidfile FILE	Create pidfile
-s,--script PROG	Run PROG at DHCP events (default /usr/share/udhcpc/default.script)
-t,--retries N	Send up to N discover packets
-T,--timeout N	Pause between packets (default 3 seconds)
-A,--tryagain N	Wait N seconds after failure (default 20)
-f,--foreground	Run in foreground
-n,--now	Exit if lease is not obtained
-q,--quit	Exit after obtaining lease
-R,--release	Release IP on exit
-S,--syslog	Log to syslog too
-P,--client-port N	Use port N (default 68)
-a,--arping	Use arping to validate offered address
-O,--request-option OPT	Request option OPT from server (cumulative)
-o,--no-default-options	Don't request any options (unless -O is given)
-r,--request IP	Request this IP address
-x OPT:VAL	Include option OPT in sent packets (cumulative)
	Examples of string, numeric, and hex byte opts:
	-x hostname:bbox - option 12
	-x lease:3600 - option 51 (lease time)
	-x 0x3d:0100BEEFC0FFEE - option 61 (client id)
-F,--fqdn NAME	Ask server to update DNS mapping for NAME
-H,-h,--hostname NAME	Send NAME as client hostname (default none)
-V,--vendorclass VENDOR	Vendor identifier (default 'udhcp VERSION')

-C, --clientid-none
-v

Don't send MAC as client identifier
Verbose

udhcpd

udhcpd [-fS] [-P N] [CONFFILE]

DHCP server

-f Run in foreground
-S Log to syslog too
-P N Use port N (default 67)

udpsvd

udpsvd [-hEv] [-c N] [-u USER] [-l NAME] IP PORT PROG

Create UDP socket, bind to IP:PORT and wait for incoming packets. Run PROG for each packet, redirecting all further packets with same peer ip:port to it.

IP IP to listen on. '0' = all
PORT Port to listen on
PROG ARGS Program to run
-l NAME Local hostname (else looks up local hostname in DNS)
-u USER[:GRP] Change to user/group after bind
-c N Handle up to N connections simultaneously
-h Look up peer's hostname
-E Don't set up environment variables
-v Verbose

umount

umount [OPTIONS] FILESYSTEM|DIRECTORY

Unmount file systems

Options:

-a Unmount all file systems in /etc/mtab
-n Don't erase /etc/mtab entries
-r Try to remount devices as read-only if mount is busy
-l Lazy unmount (detach filesystem)
-f Force unmount (i.e., unreachable NFS server)
-d Free loop device if it has been used

uname

uname [-amnrspv]

Print system information

Options:

-a Print all
-m The machine (hardware) type
-n Hostname
-r OS release
-s OS name (default)
-p Processor type
-v OS version

uncompress

uncompress [-cf] [FILE]...

Decompress .Z file[s]

Options:

-c Write to stdout
-f Overwrite

unexpand

unexpand [-fa][-t N] [FILE]...

Convert spaces to tabs, writing to stdout

Options:

-a,--all	Convert all blanks
-f,--first-only	Convert only leading blanks
-t,--tabs=N	Tabstops every N chars

uniq

uniq [-cdu][-f,s,w N] [INPUT [OUTPUT]]

Discard duplicate lines

Options:

-c	Prefix lines by the number of occurrences
-d	Only print duplicate lines
-u	Only print unique lines
-f N	Skip first N fields
-s N	Skip first N chars (after any skipped fields)
-w N	Compare N characters in line

unix2dos

unix2dos [-ud] [FILE]

Convert FILE in-place from Unix to DOS format. When no file is given, use stdin/stdout.

Options:

-u	dos2unix
-d	unix2dos

unlzma

unlzma [-cf] [FILE]...

Decompress FILE (or stdin)

Options:

-c	Write to stdout
-f	Force

unlzop

unlzop [-cfvCF] [FILE]...

Options:

-c	Write to stdout
-f	Force
-v	Verbose
-F	Don't store or verify checksum

unxz

unxz [-cf] [FILE]...

Decompress FILE (or stdin)

Options:

-c	Write to stdout
-f	Force

unzip

unzip [-opts[modifiers]] FILE[.zip] [LIST] [-x XLIST] [-d DIR]

Extract files from ZIP archives

Options:

```
-l      List archive contents (with -q for short form)
-n      Never overwrite files (default)
-o      Overwrite
-p      Send output to stdout
-q      Quiet
-x XLST Exclude these files
-d DIR  Extract files into DIR
```

uptime

uptime

Display the time since the last boot

usleep

usleep N

Pause for N microseconds

uudecode

uudecode [-o OUTFILE] [INFILE]

Udecode a file Finds outfile name in uuencoded source unless -o is given

uuencode

uuencode [-m] [INFILE] STORED_FILENAME

Uuencode a file to stdout

Options:

```
-m      Use base64 encoding per RFC1521
```

vconfig

vconfig COMMAND [OPTIONS]

Create and remove virtual ethernet devices

Options:

```
add      [interface-name] [vlan_id]
rem      [vlan-name]
set_flag [interface-name] [flag-num] [0 | 1]
set_egress_map [vlan-name] [skb_priority] [vlan_qos]
set_ingress_map [vlan-name] [skb_priority] [vlan_qos]
set_name_type [name-type]
```

vi

vi [OPTIONS] [FILE]...

Edit FILE

Options:

```
-c      Initial command to run ($EXINIT also available)
-R      Read-only
-H      Short help regarding available features
```

vlock

vlock [-a]

Lock a virtual terminal. A password is required to unlock.

Options:

-a Lock all VTs

volname

volname [DEVICE]

Show CD volume name of the DEVICE (default /dev/cdrom)

wall

wall [FILE]

Write content of FILE or stdin to all logged-in users

watch

watch [-n SEC] [-t] PROG ARGS

Run PROG periodically

Options:

-n Loop period in seconds (default 2)
-t Don't print header

watchdog

watchdog [-t N[ms]] [-T N[ms]] [-F] DEV

Periodically write to watchdog device DEV

Options:

-T N Reboot after N seconds if not reset (default 60)
-t N Reset every N seconds (default 30)
-F Run in foreground

Use 500ms to specify period in milliseconds

wc

wc [-cmlwL] [FILE]...

Count lines, words, and bytes for each FILE (or stdin)

Options:

-c Count bytes
-m Count characters
-l Count newlines
-w Count words
-L Print longest line length

wget

wget [-c|--continue] [-s|--spider] [-q|--quiet] [-O|--output-document FILE]

[--header 'header: value'] [-Y|--proxy on/off] [-P DIR]
[--no-check-certificate] [-U|--user-agent AGENT][--T SEC] URL

Retrieve files via HTTP or FTP

Options:

-s Spider mode - only check file existence
-c Continue retrieval of aborted transfer
-q Quiet
-P DIR Save to DIR (default .)
-T SEC Network read timeout is SEC seconds

```
-O FILE Save to FILE ('-' for stdout)
-U STR  Use STR for User-Agent header
-Y      Use proxy ('on' or 'off')
```

which

which [COMMAND]...

Locate a COMMAND

who

who [-a]

Show who is logged on

Options:

```
-a      Show all
```

whoami

whoami

Print the user name associated with the current effective user id

xargs

xargs [OPTIONS] [PROG ARGS]

Run PROG on every item given by stdin

Options:

```
-p      Ask user whether to run each command
-r      Don't run command if input is empty
-0      Input is separated by NUL characters
-t      Print the command on stderr before execution
-e[STR] STR stops input processing
-n N    Pass no more than N args to PROG
-s N    Pass command line of no more than N bytes
-x      Exit if size is exceeded
```

xz

xz -d [-cf] [FILE]...

Decompress FILE (or stdin)

Options:

```
-d      Decompress
-c      Write to stdout
-f      Force
```

xzcat

xzcat FILE

Decompress to stdout

yes

yes [STRING]

Repeatedly output a line with STRING, or 'y'

zcat

zcat FILE

Decompress to stdout

zcip

zcip [OPTIONS] IFACE SCRIPT

Manage a ZeroConf IPv4 link-local address

Options:

-f	Run in foreground
-q	Quit after obtaining address
-r 169.254.x.x	Request this address first
-v	Verbose

With no -q, runs continuously monitoring for ARP conflicts, exits only on I/O errors (link down etc)

LIBC NSS

GNU Libc (glibc) uses the Name Service Switch (NSS) to configure the behavior of the C library for the local environment, and to configure how it reads system data, such as passwords and group information. This is implemented using an `/etc/nsswitch.conf` configuration file, and using one or more of the `/lib/libnss_*` libraries. BusyBox tries to avoid using any libc calls that make use of NSS. Some applets however, such as `login` and `su`, will use libc functions that require NSS.

If you enable `CONFIG_USE_BB_PWD_GRP`, BusyBox will use internal functions to directly access the `/etc/passwd`, `/etc/group`, and `/etc/shadow` files without using NSS. This may allow you to run your system without the need for installing any of the NSS configuration files and libraries.

When used with glibc, the BusyBox 'networking' applets will similarly require that you install at least some of the glibc NSS stuff (in particular, `/etc/nsswitch.conf`, `/lib/libnss_dns*`, `/lib/libnss_files*`, and `/lib/libresolv*`).

Shameless Plug: As an alternative, one could use a C library such as uClibc. In addition to making your system significantly smaller, uClibc does not require the use of any NSS support files or libraries.

MAINTAINER

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AUTHORS

The following people have contributed code to BusyBox whether they know it or not. If you have written code included in BusyBox, you should probably be listed here so you can obtain your bit of eternal glory. If you should be listed here, or the description of what you have done needs more detail, or is incorrect, please send in an update.

Emanuele Aina <emanuele.aina@tiscali.it> run-parts

Erik Andersen <andersen@codepoet.org>

Tons of new stuff, major rewrite of most of the core apps, tons of new apps as noted in header files. Lots of tedious effort writing these boring docs that nobody is going to actually read.

Laurence Anderson <l.d.anderson@warwick.ac.uk>

rpm2cpio, unzip, get_header_cpio, read_gz interface, rpm

Jeff Angielski <jeff@theptgroup.com>

ftpput, ftpget

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expr, hostid, logname, whoami

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du, nslookup, sort

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Dave Cinege <dcinege@psychosis.com>

more(v2), makedevs, dutmp, modularization, auto links file,
various fixes, Linux Router Project maintenance

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ipcalc

Magnus Damm <damm@opensource.se>

tftp client insmod powerpc support

Larry Doolittle <ldoolitt@recycle.lbl.gov>

pristine source directory compilation, lots of patches and fixes.

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httpd

Gennady Feldman <gfeldman@gena01.com>

Sysklogd (single threaded syslogd, IPC Circular buffer support,
logread), various fixes.

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cp_mv.c, the test suite, various fixes to utility.c, &c.

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mktemp.c

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documentation, bugfixes, test suite

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ipcalc, Red Hat equivalence

John Lombardo <john@deltanet.com>

tr

Glenn McGrath <bug1@iinet.net.au>

Common unarchiving code and unarchiving applets, ifupdown, ftpgetput, nameif, sed, patch, fold, install, uudecode.
Various bugfixes, review and apply numerous patches.

Manuel Novoa III <mjn3@codepoet.org>

cat, head, mkfifo, mknod, rmdir, sleep, tee, tty, uniq, usleep, wc, yes, msg, vconfig, make_directory, parse_mode, dirname, mode_string, get_last_path_component, simplify_path, and a number trivial libbb routines

also bug fixes, partial rewrites, and size optimizations in ash, basename, cal, cmp, cp, df, du, echo, env, ln, logname, md5sum, mkdir, mv, realpath, rm, sort, tail, touch, uname, watch, arith, human_readable, interface, dumpt, ifconfig, route

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cmdedit; xargs(current), httpd(current);
ports: ash, crond, fdisk, inetd, stty, traceroute, top;
locale, various fixes
and irreconcilable critic of everything not perfect.

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Original author of BusyBox in 1995, 1996. Some of his code can still be found hiding here and there...

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bug fixes, member of fan club

Kent Robotti <robotti@metconnect.com>

reset, tons and tons of bug reports and patches.

Chip Rosenthal <chip@unicom.com>, <crosth@covad.com>

wget - Contributed by permission of Covad Communications

Pavel Roskin <proski@gnu.org>

Lots of bugs fixes and patches.

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Remote logging feature for syslogd

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mkswap, fsck.minix, mkfs.minix

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grep, sed, cut, xargs(previous),
style-guide, new-applet-HOWTO, bug fixes, etc.

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gzip, mini-netcat(nc)

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tarcat (since removed), loadkmap, various fixes, Debian maintenance

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devfsd and size optimizations in strings, openvt and deallocvt.

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vi editing mode for ash, various other patches/fixes

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misc

Mike Frysinger <vapier@gentoo.org>

initial e2fsprogs, printenv, setarch, sum, misc

Jie Zhang <jie.zhang@analog.com>

fixed two bugs in msh and hush (exitcode of killed processes)